

BFS Traversal

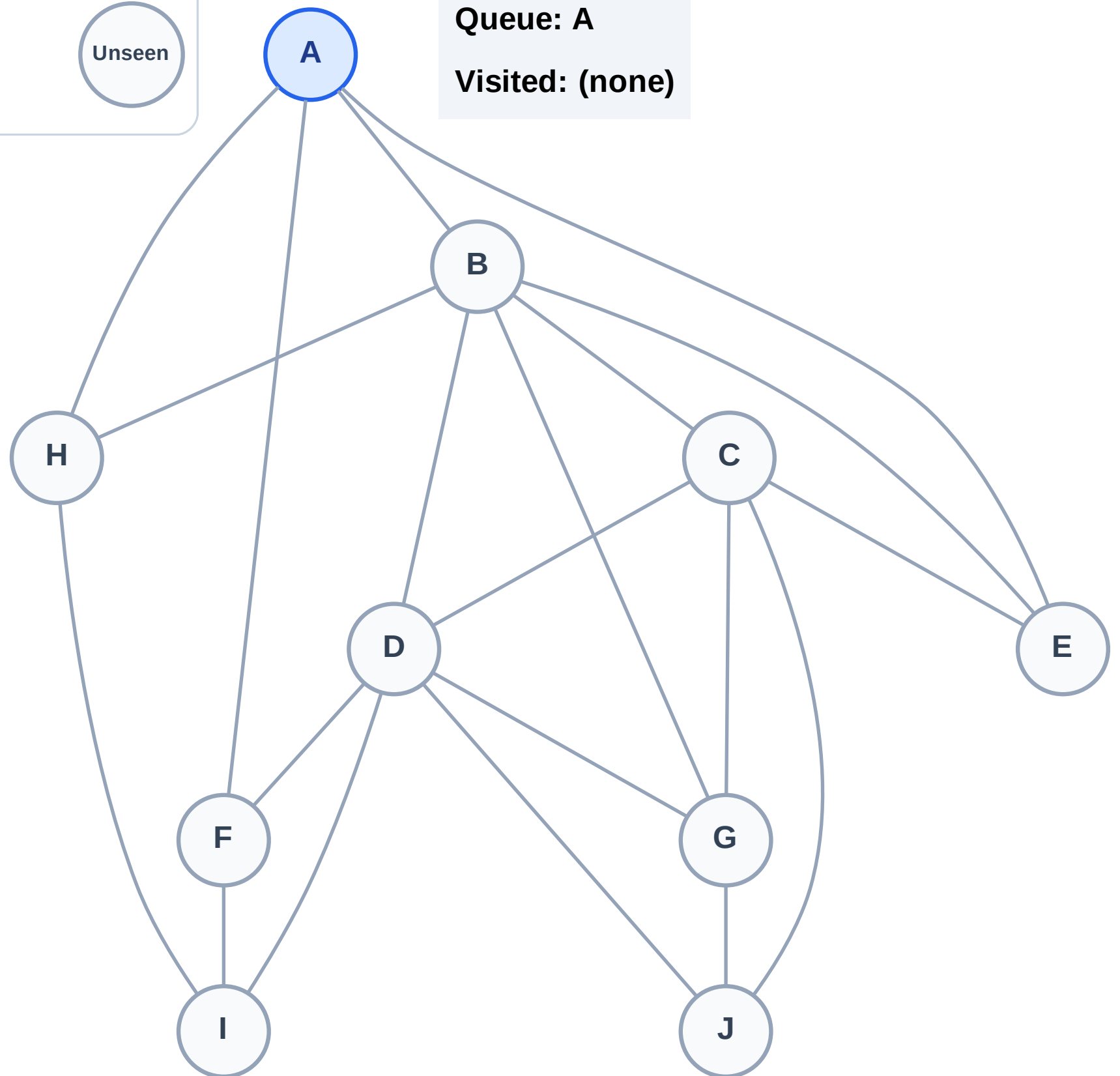
Step 1: Start at A

Legend



Queue: A

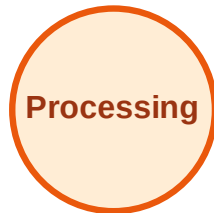
Visited: (none)



BFS Traversal

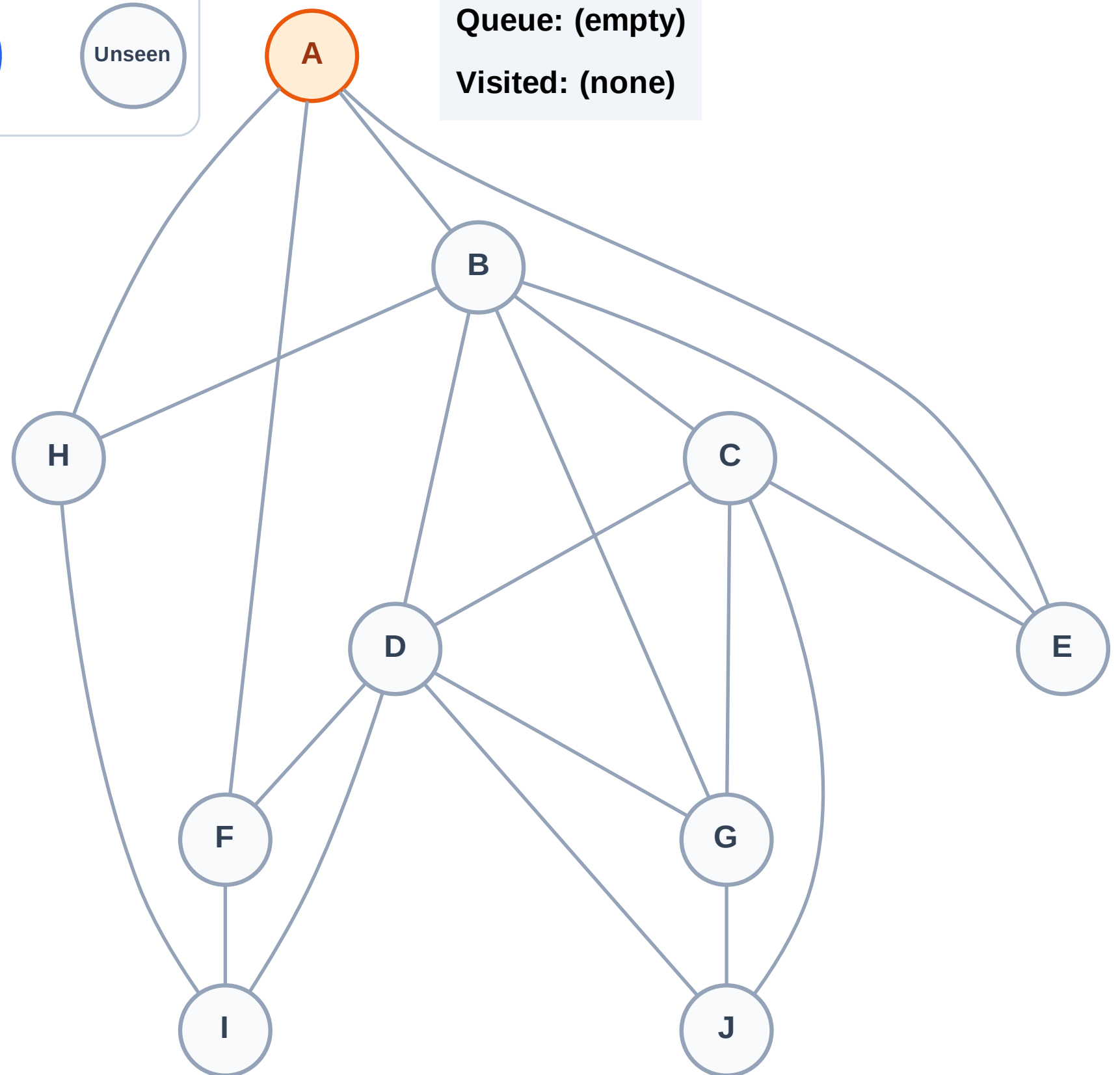
Step 2: Process node A

Legend



Queue: (empty)

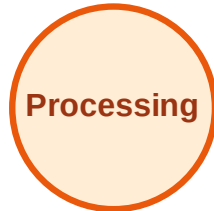
Visited: (none)



BFS Traversal

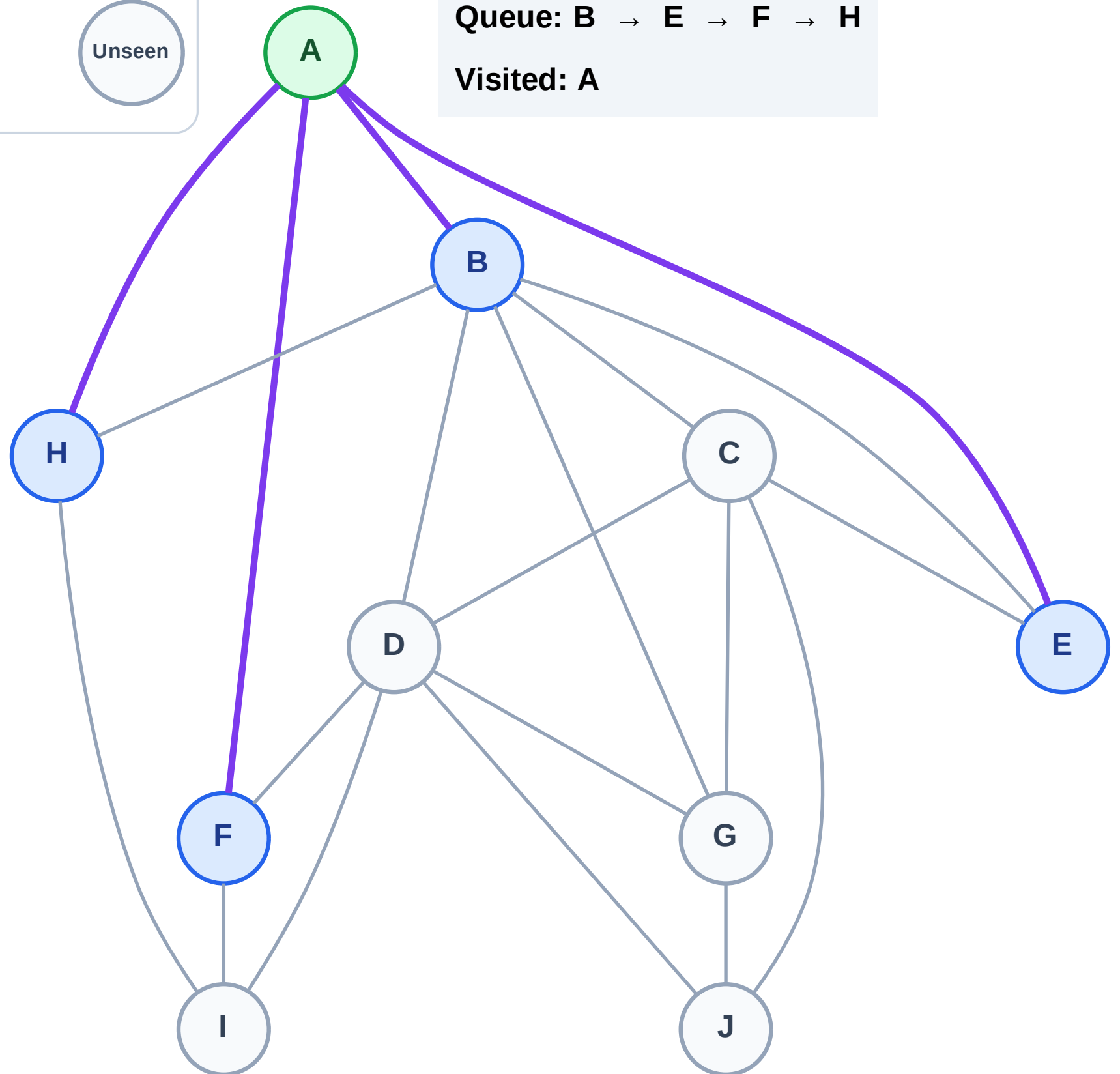
Step 3: Discover B, E, F, H from A

Legend



Queue: B → E → F → H

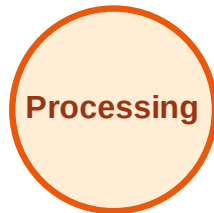
Visited: A



BFS Traversal

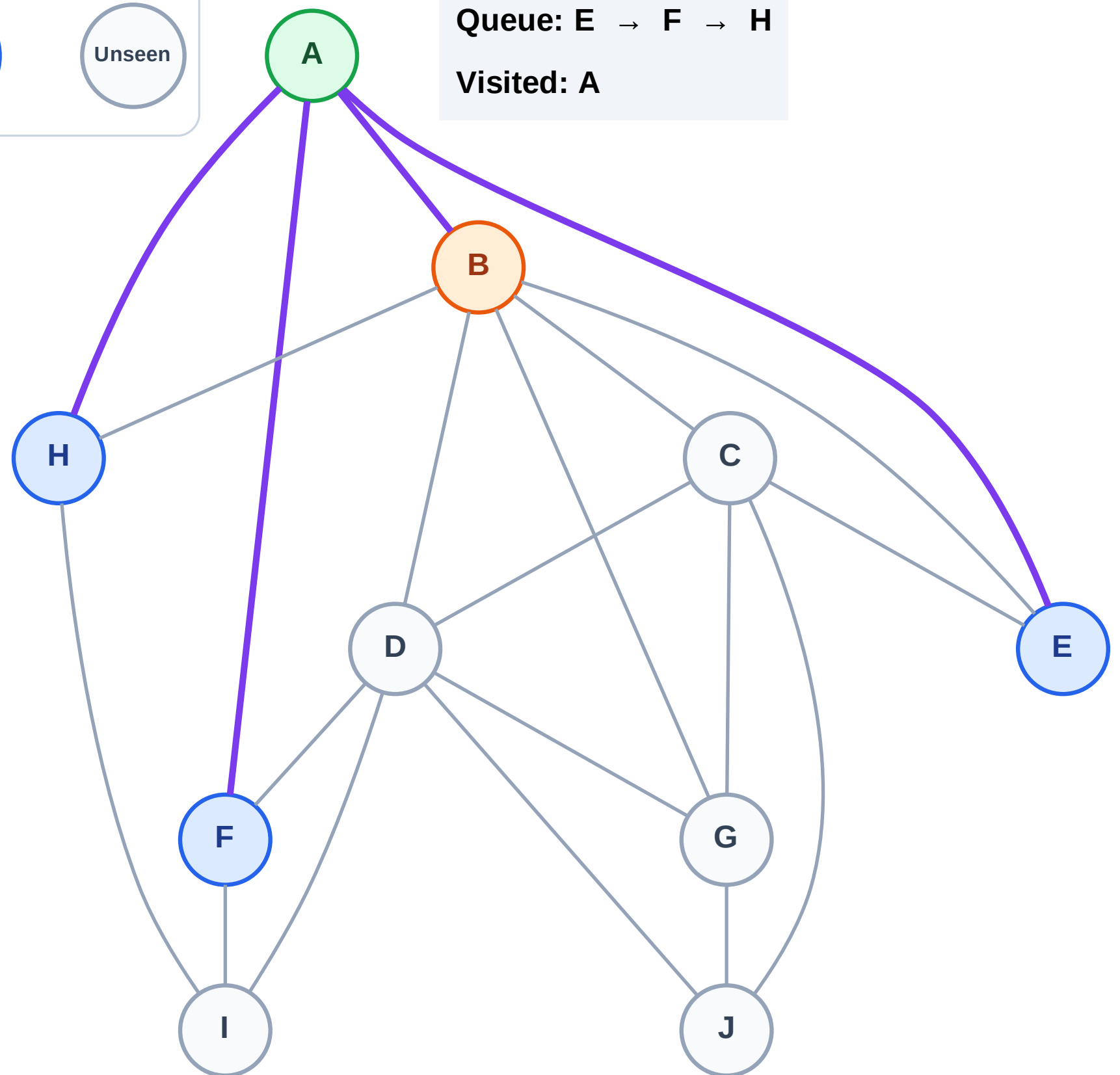
Step 4: Process node B

Legend



Queue: E → F → H

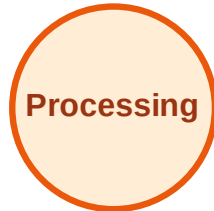
Visited: A



BFS Traversal

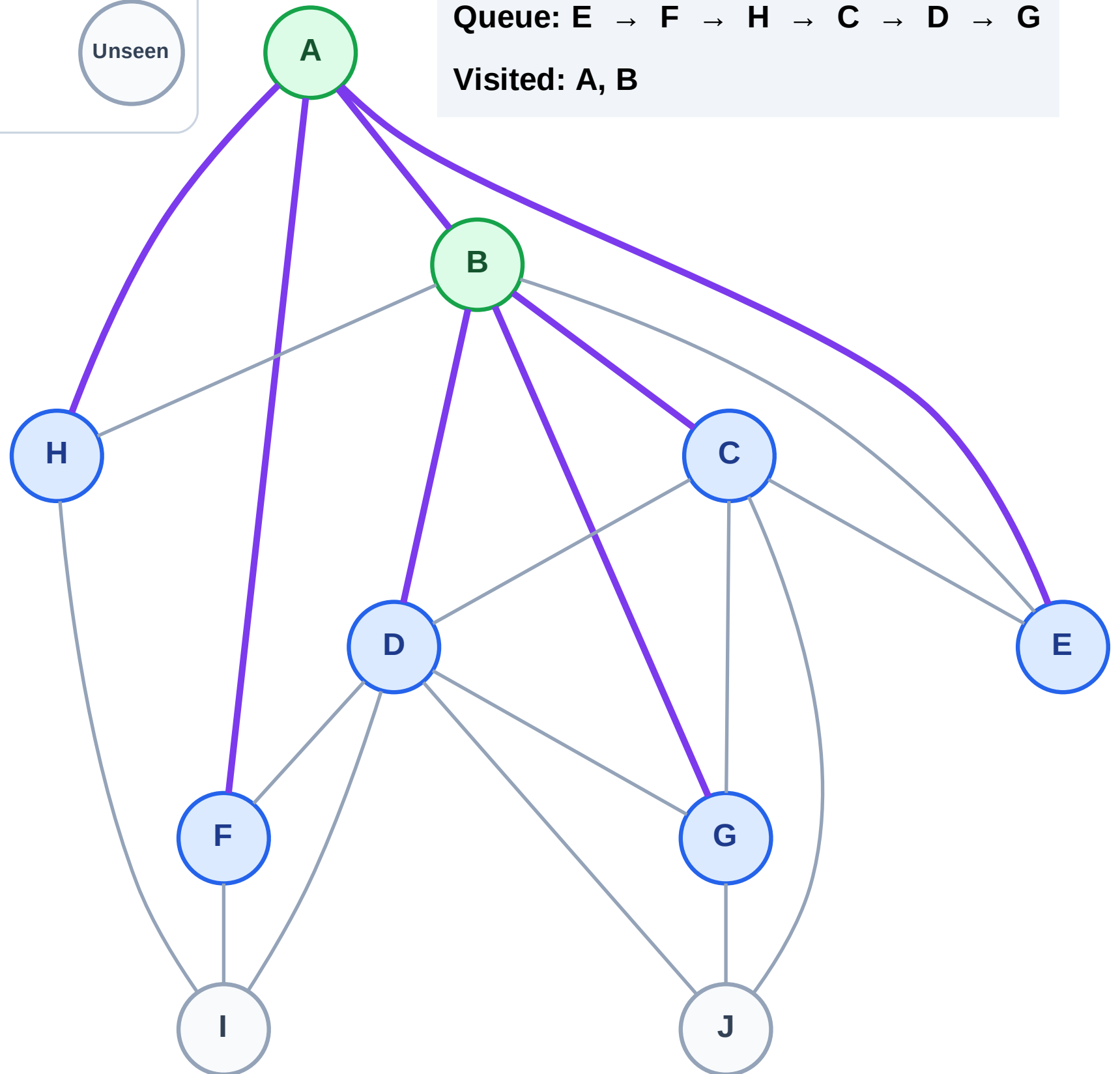
Step 5: Discover C, D, G from B

Legend



Queue: E → F → H → C → D → G

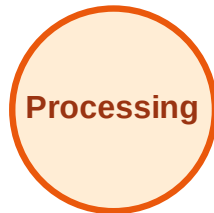
Visited: A, B



BFS Traversal

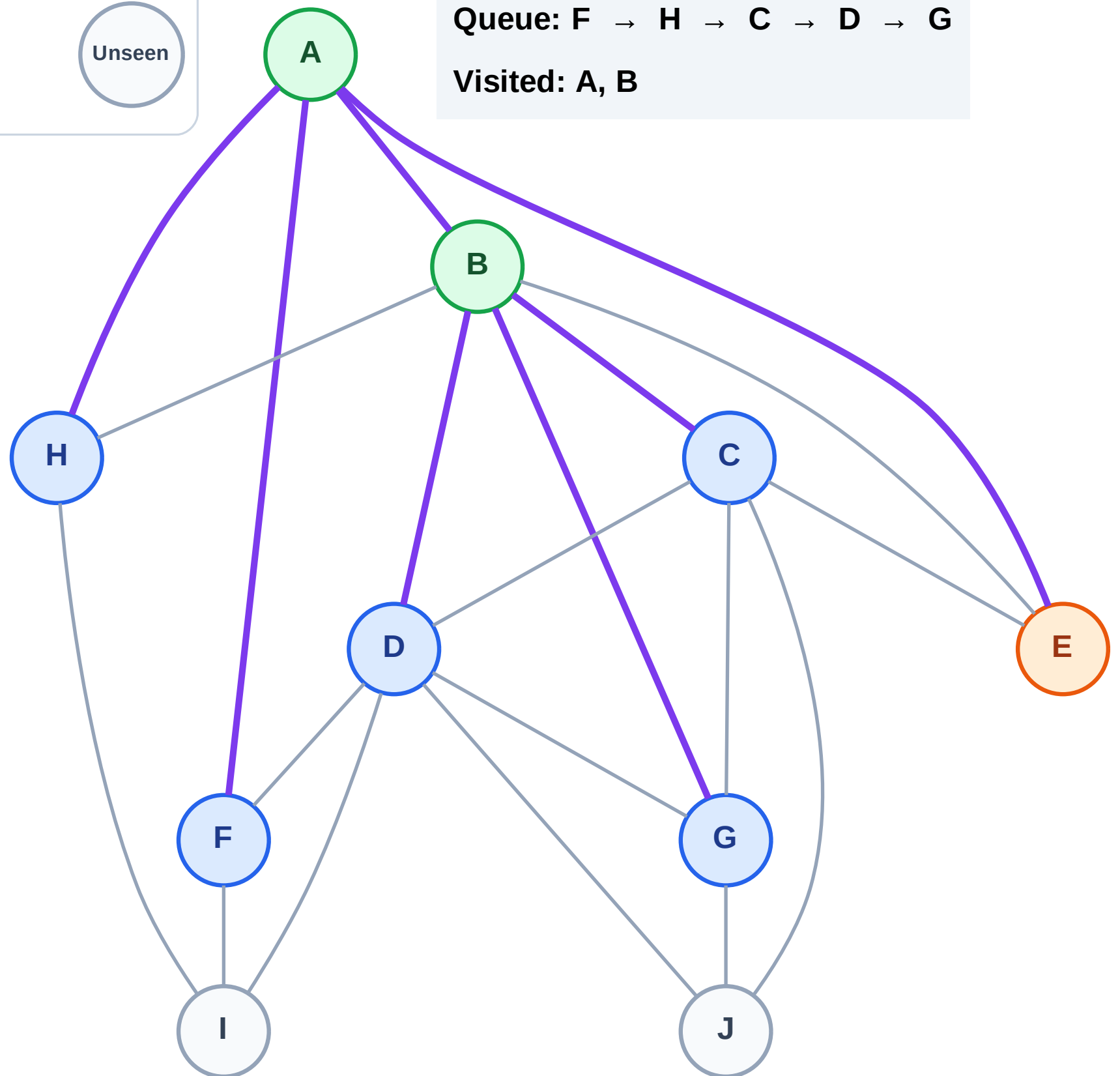
Step 6: Process node E

Legend



Queue: F → H → C → D → G

Visited: A, B



BFS Traversal

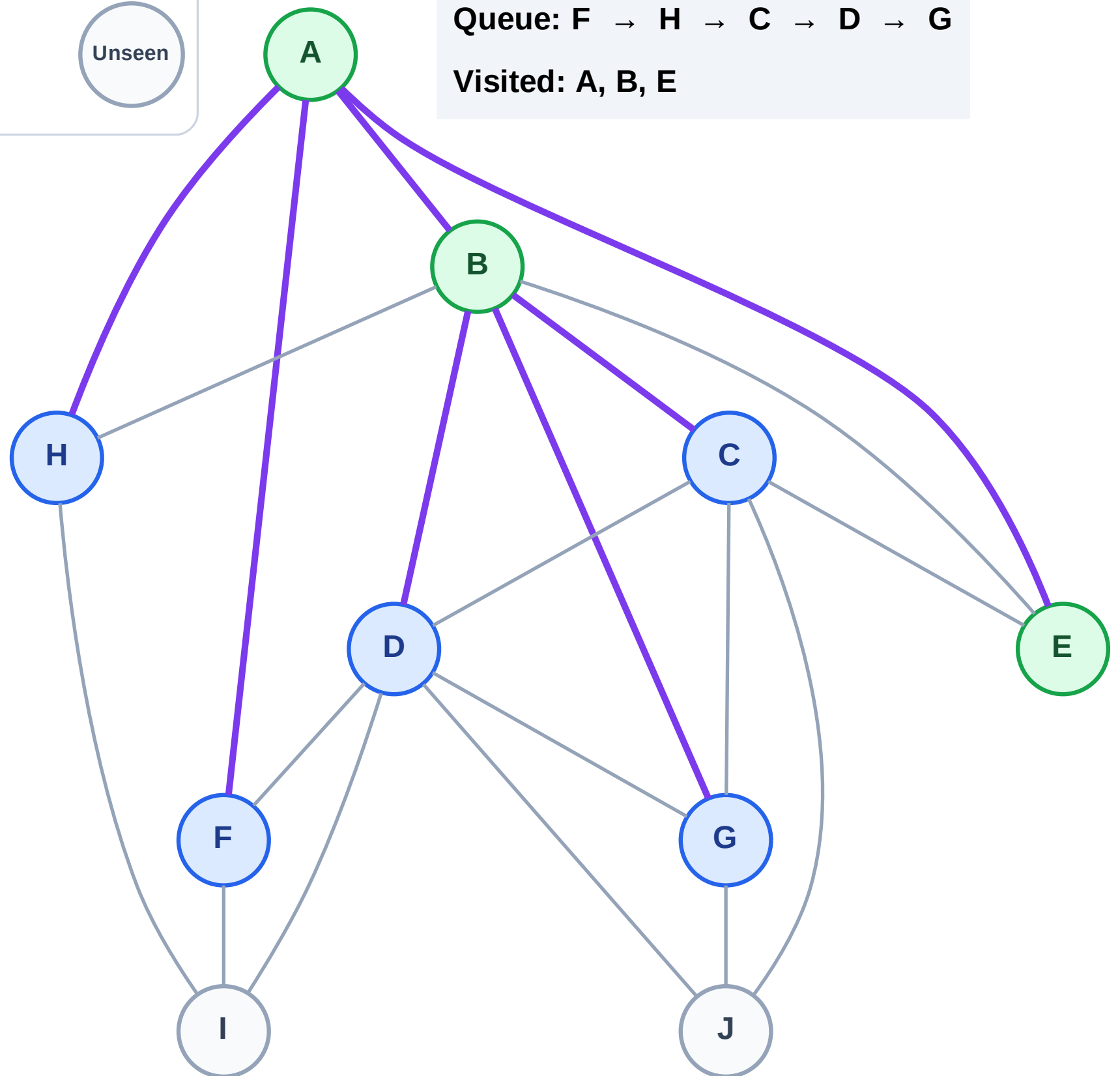
Step 7: No new neighbors from E

Legend



Queue: F → H → C → D → G

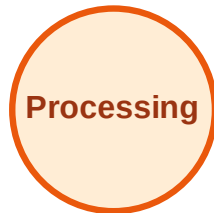
Visited: A, B, E



BFS Traversal

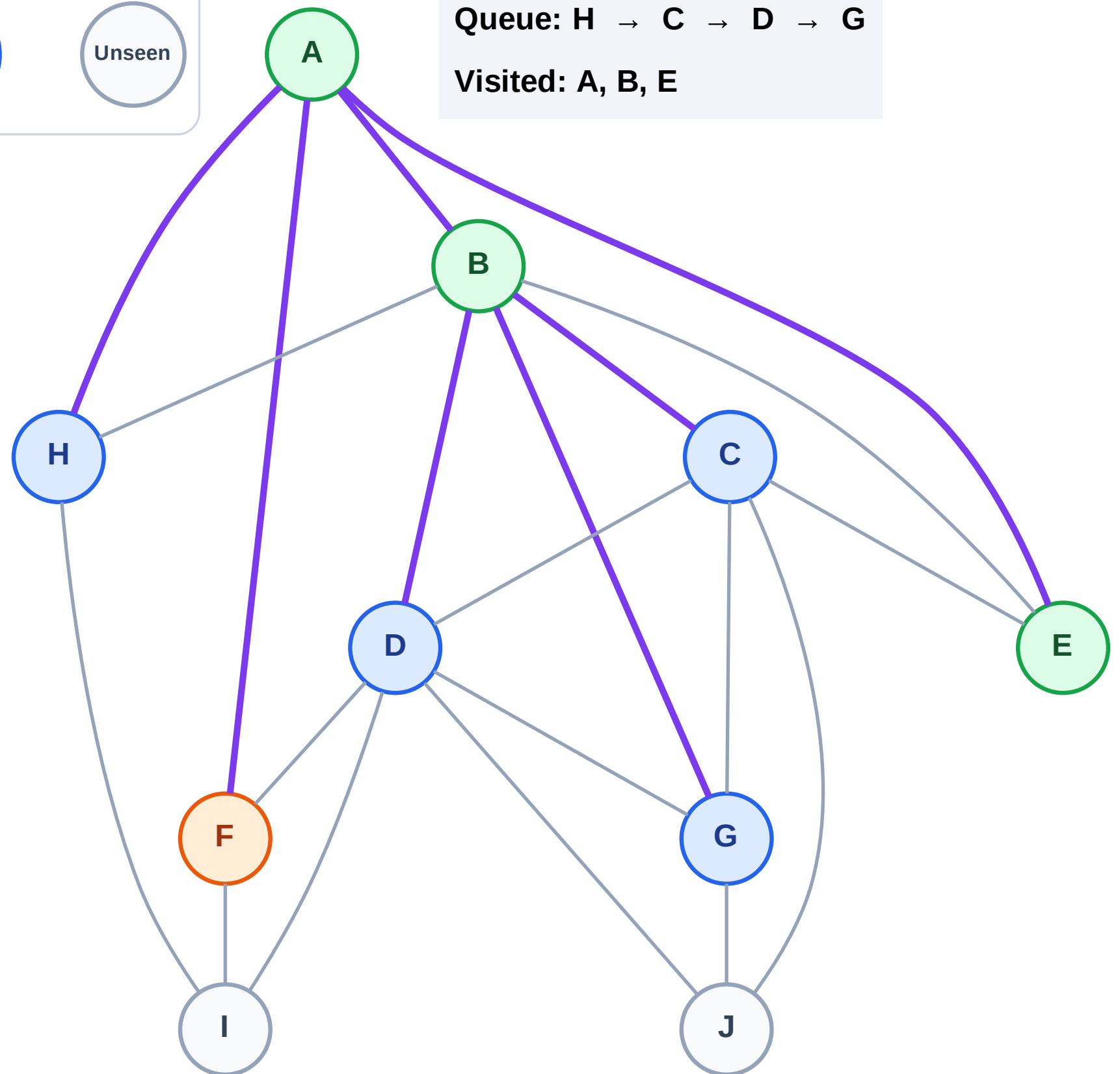
Step 8: Process node F

Legend



Queue: H → C → D → G

Visited: A, B, E



BFS Traversal

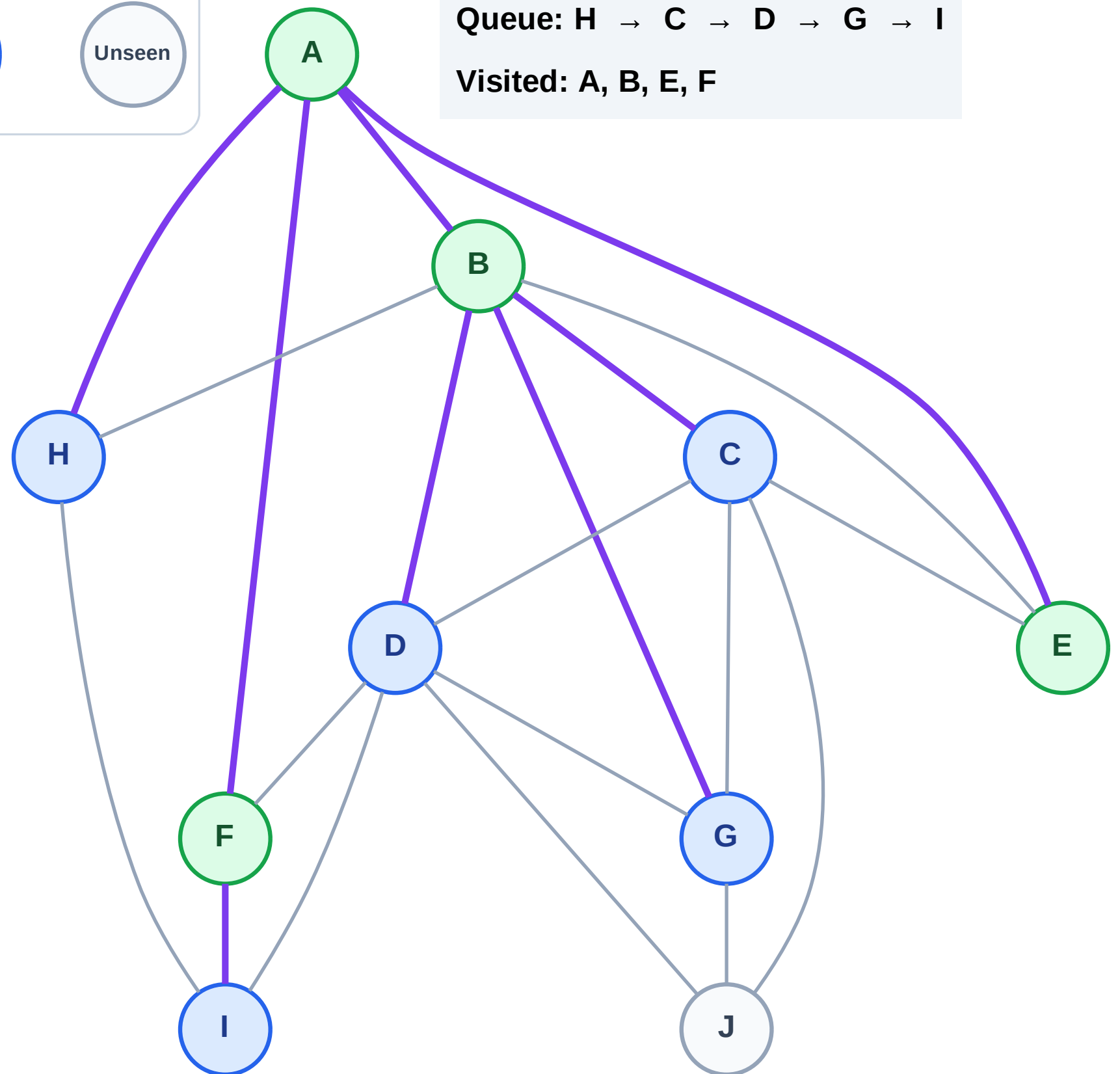
Step 9: Discover I from F

Legend



Queue: H → C → D → G → I

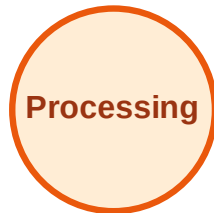
Visited: A, B, E, F



BFS Traversal

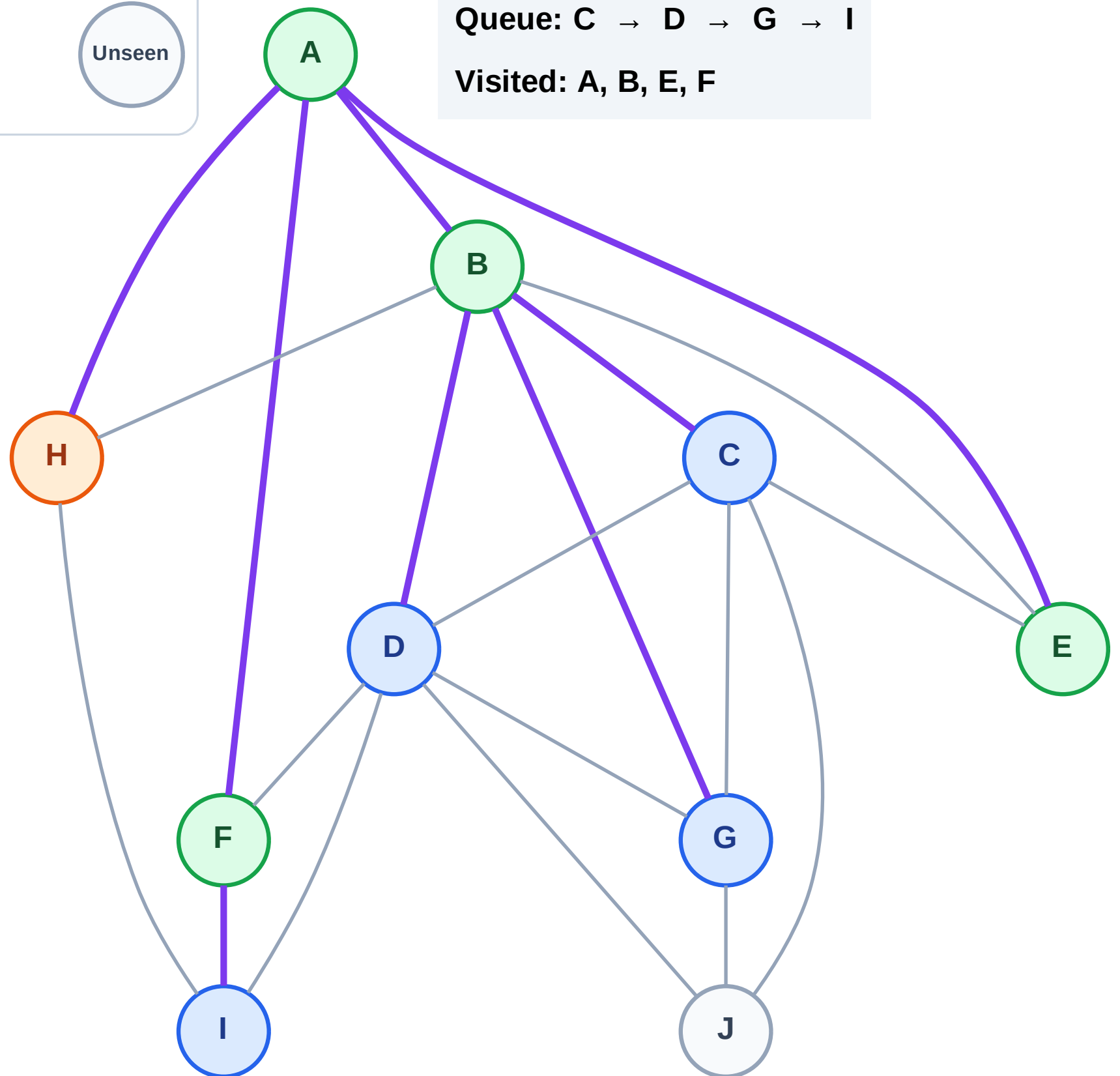
Step 10: Process node H

Legend



Queue: C → D → G → I

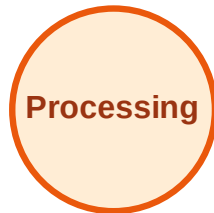
Visited: A, B, E, F



BFS Traversal

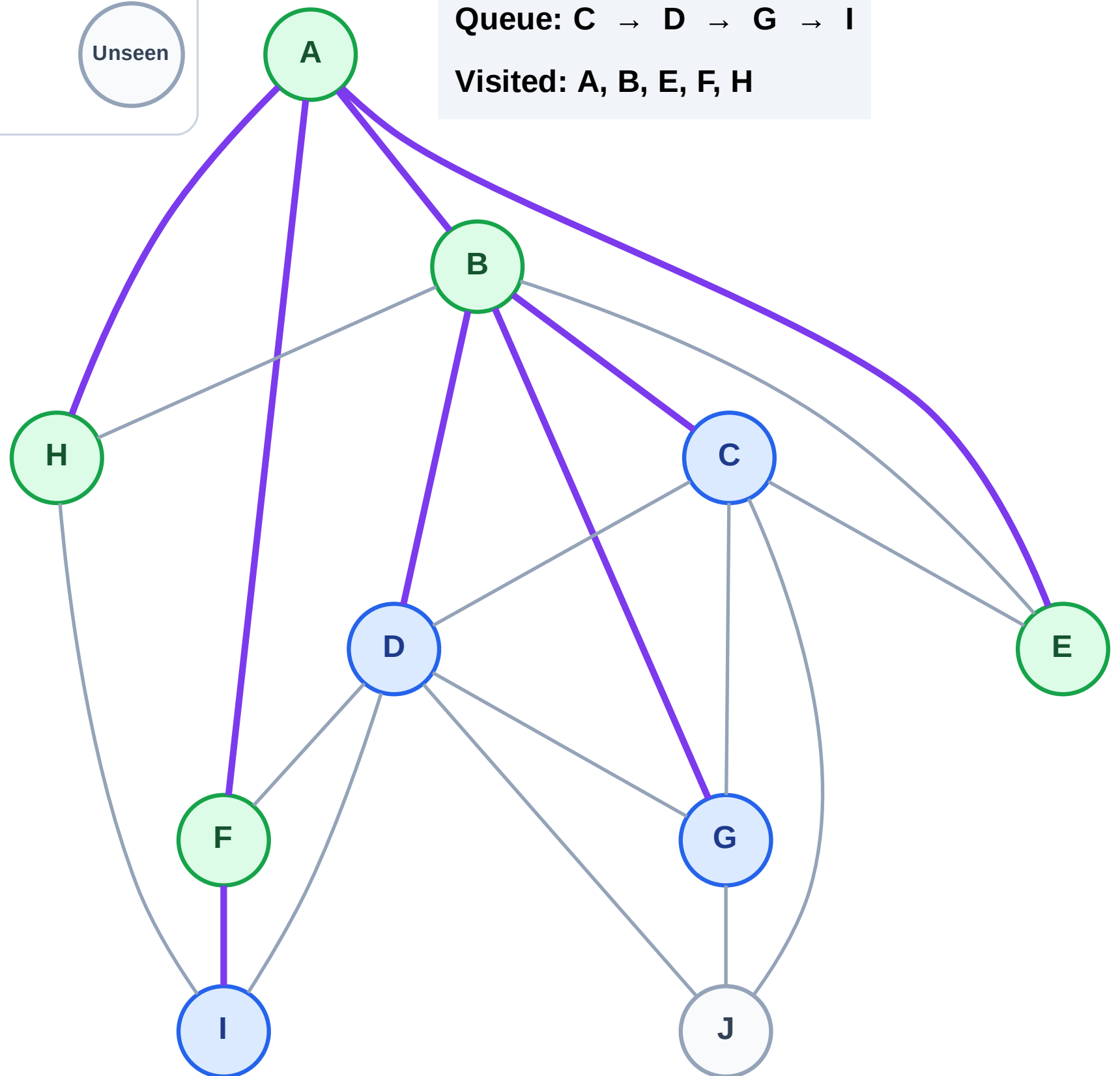
Step 11: No new neighbors from H

Legend



Queue: C → D → G → I

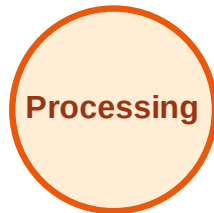
Visited: A, B, E, F, H



BFS Traversal

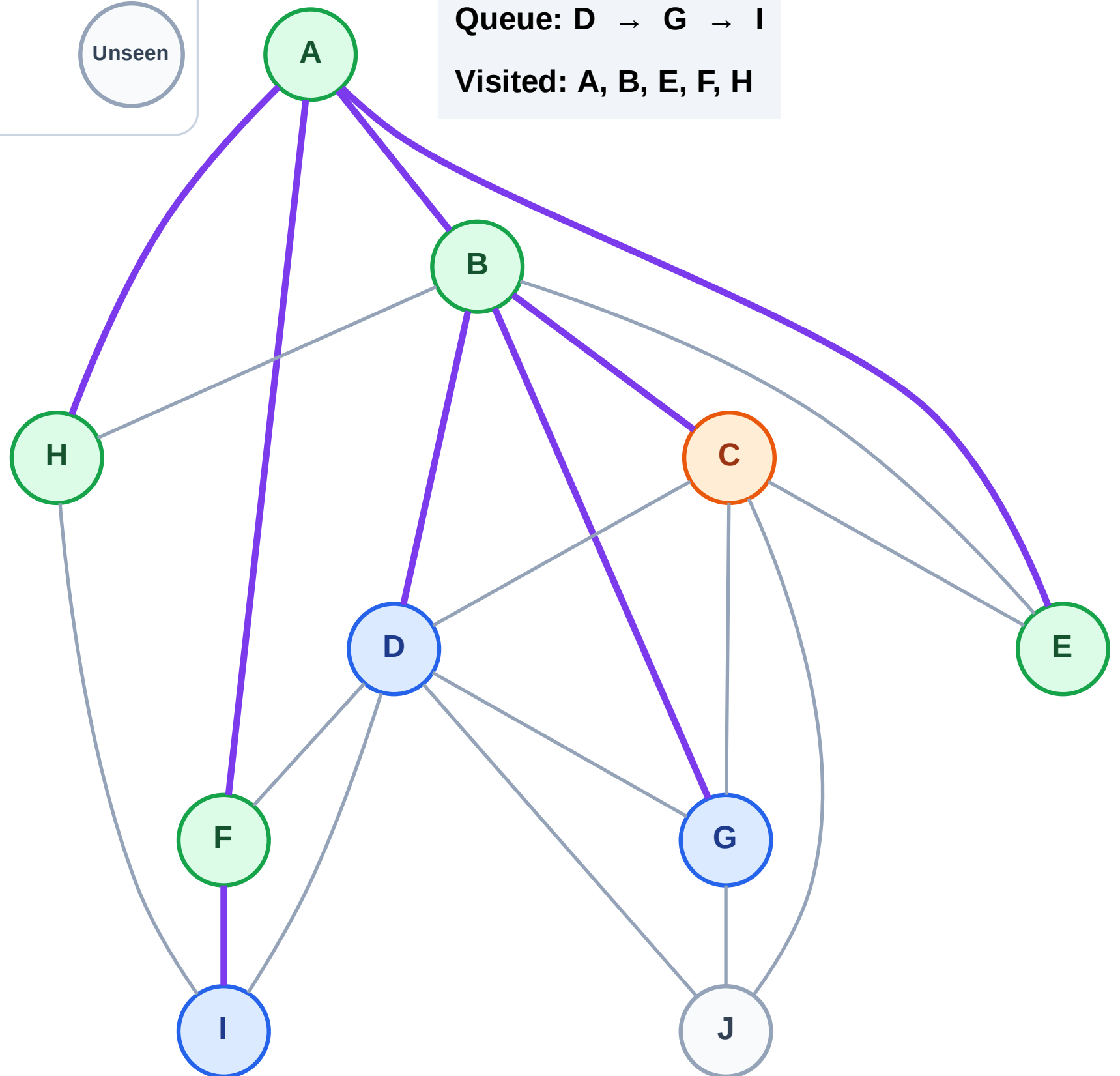
Step 12: Process node C

Legend



Queue: D → G → I

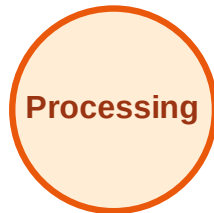
Visited: A, B, E, F, H



BFS Traversal

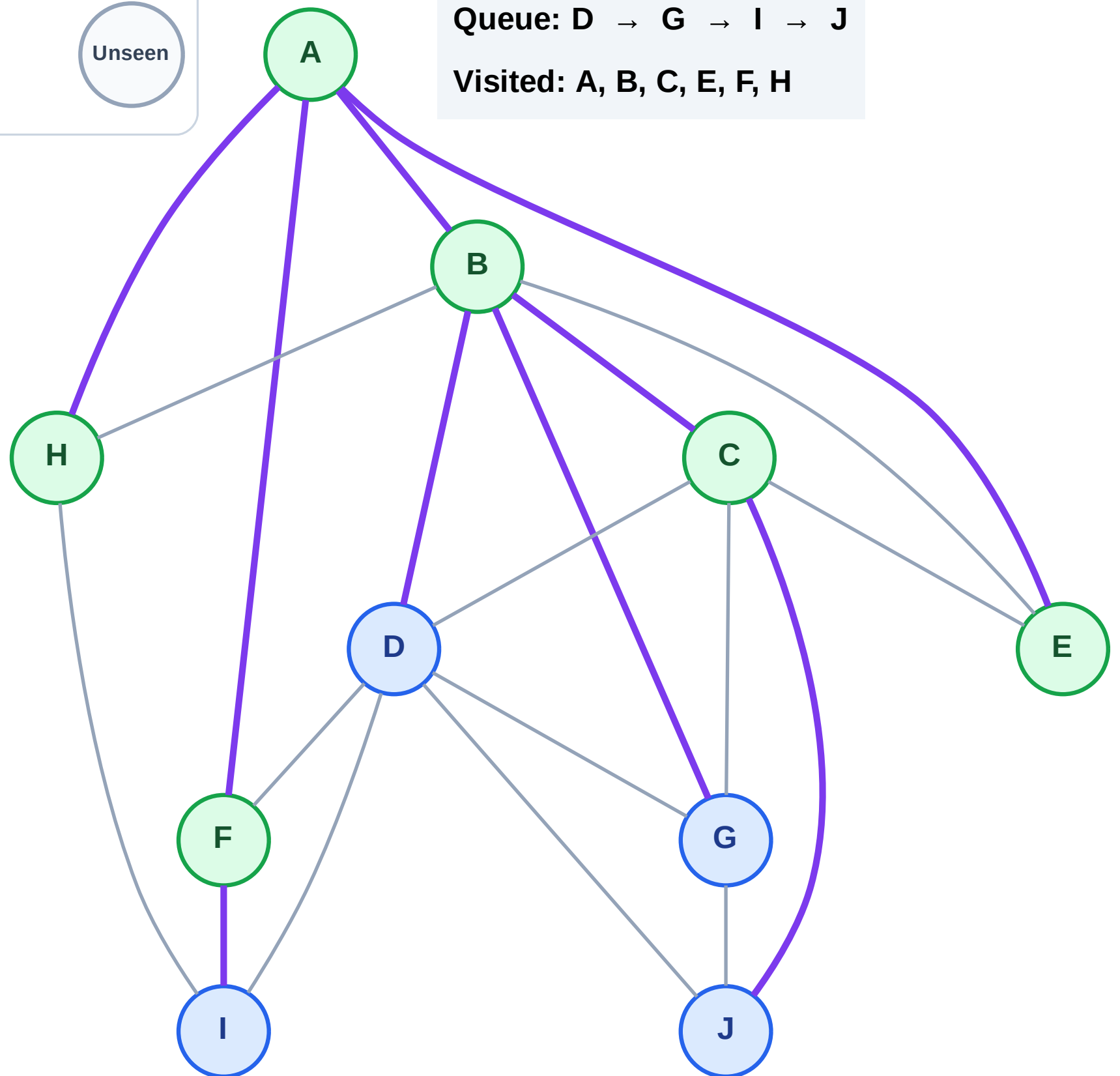
Step 13: Discover J from C

Legend



Queue: D → G → I → J

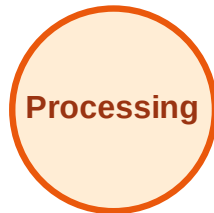
Visited: A, B, C, E, F, H



BFS Traversal

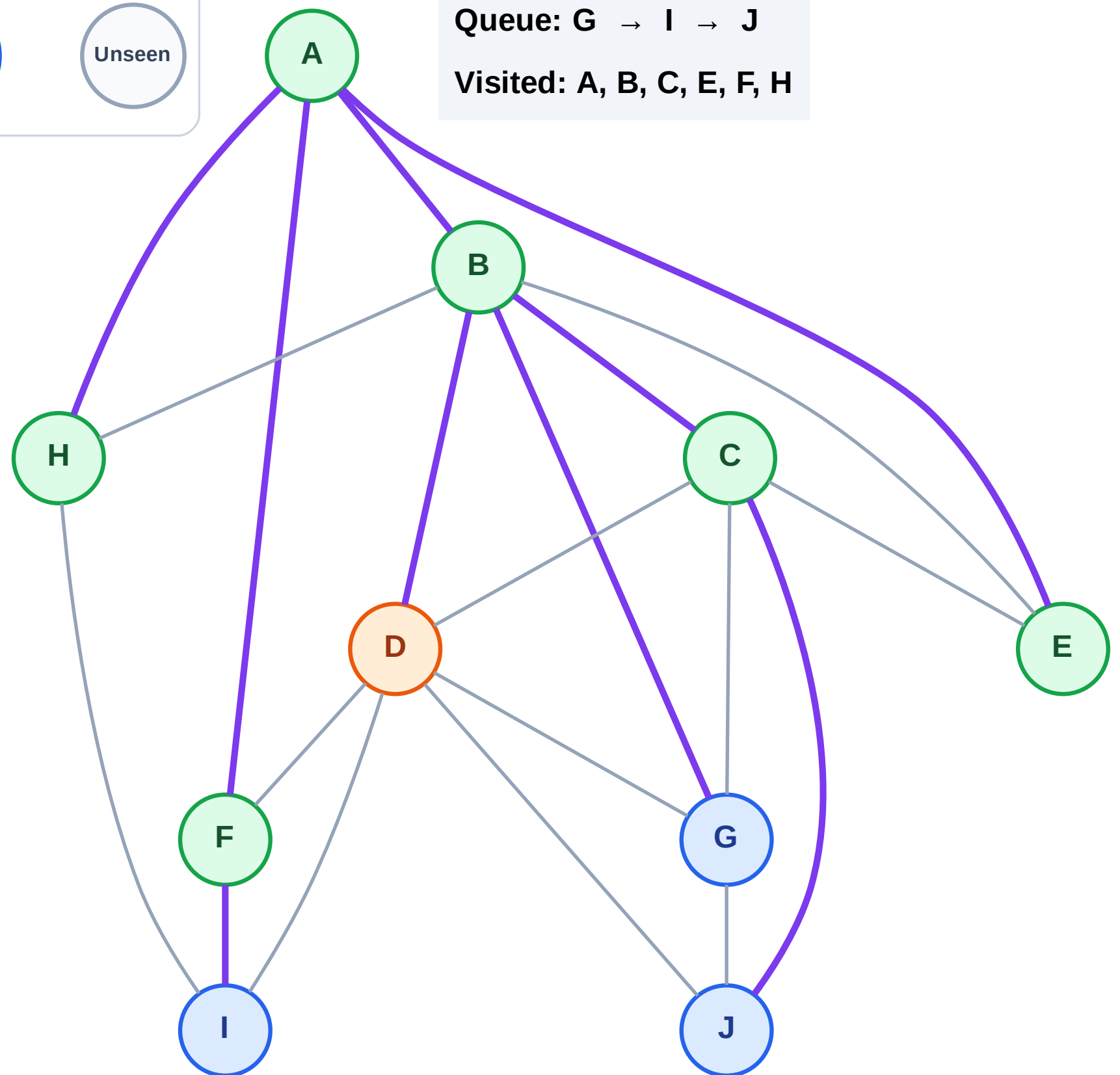
Step 14: Process node D

Legend



Queue: G → I → J

Visited: A, B, C, E, F, H



BFS Traversal

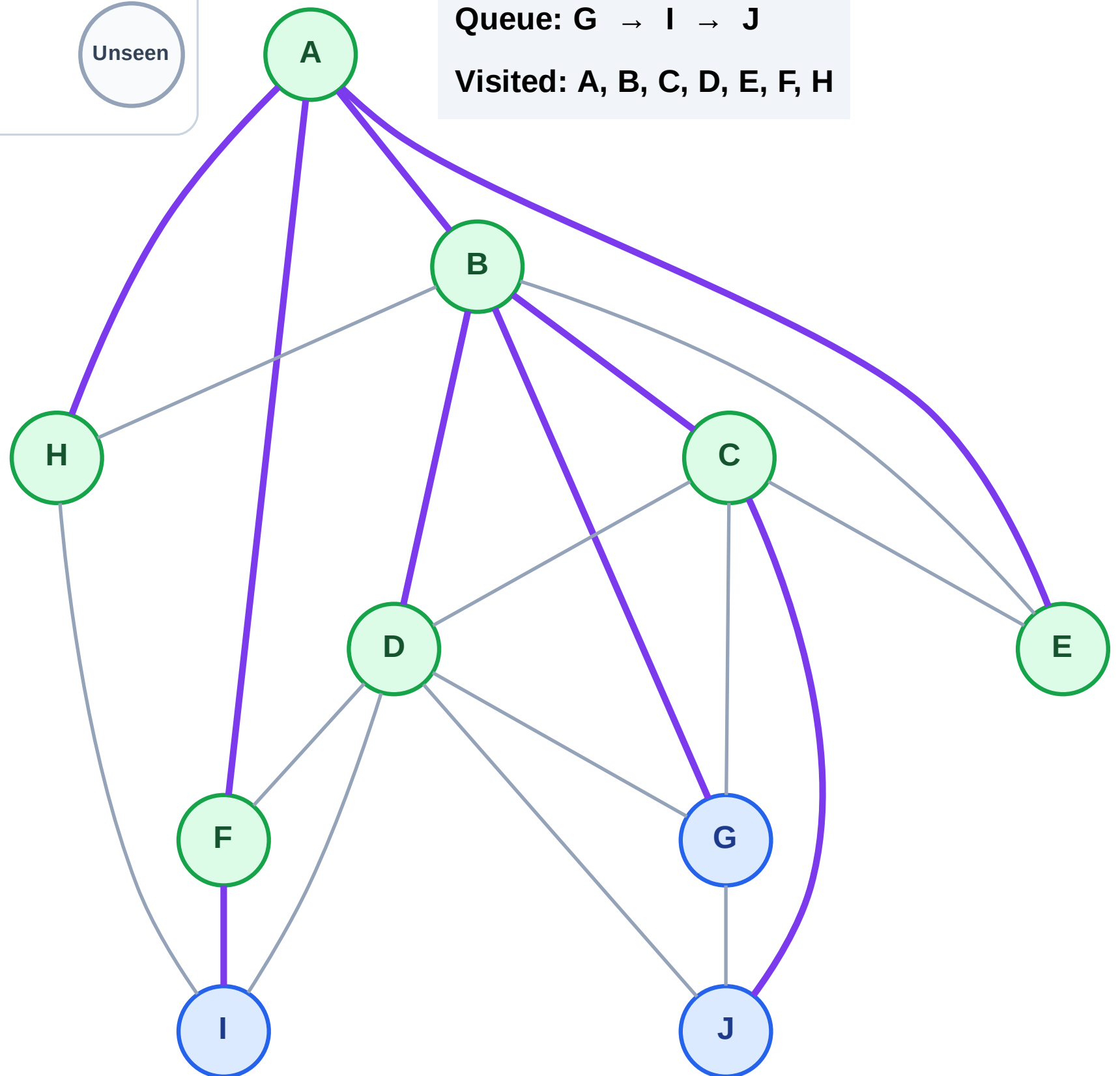
Step 15: No new neighbors from D

Legend



Queue: G → I → J

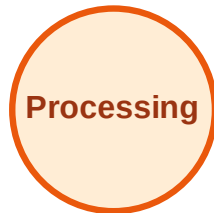
Visited: A, B, C, D, E, F, H



BFS Traversal

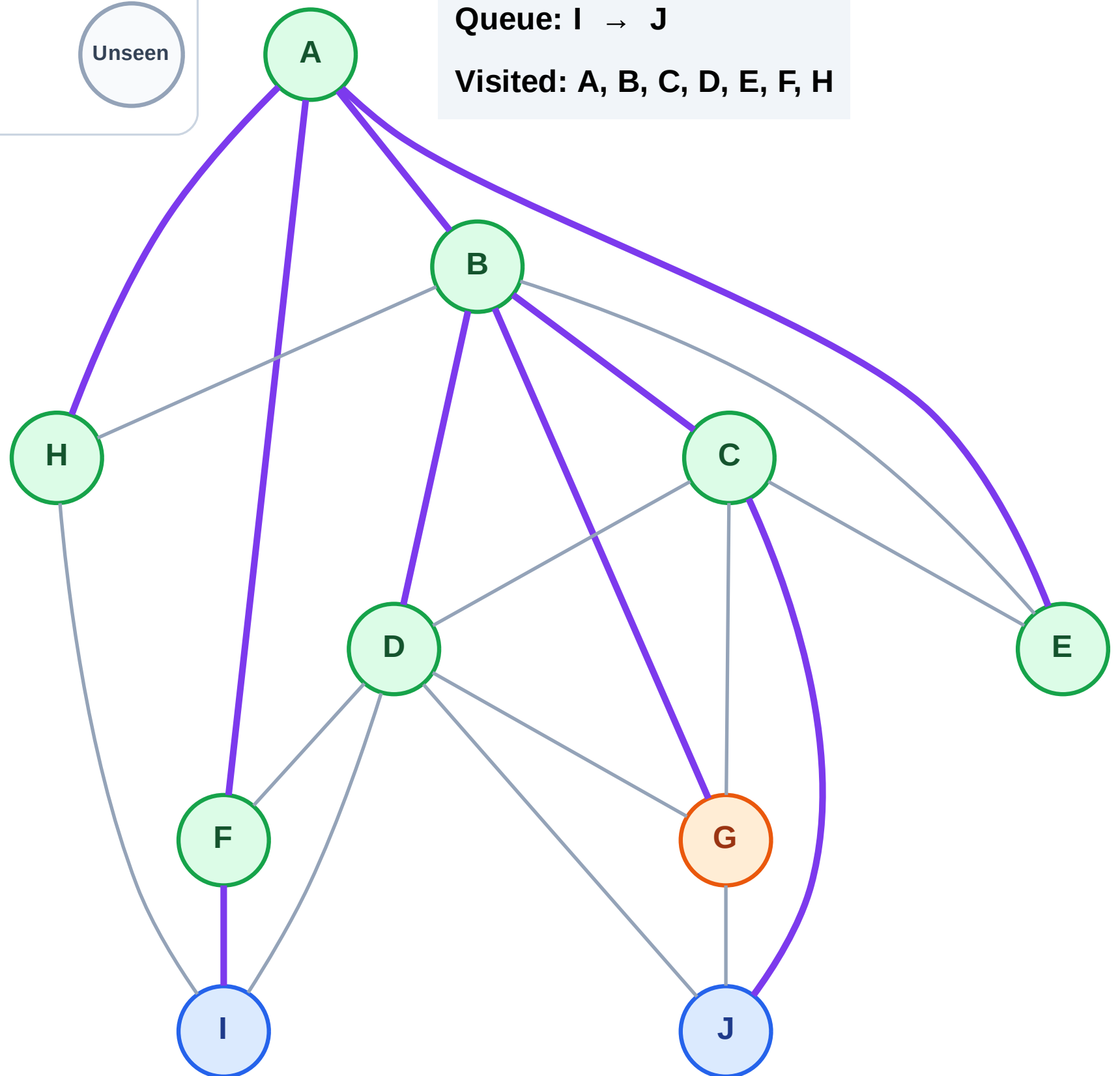
Step 16: Process node G

Legend



Queue: I → J

Visited: A, B, C, D, E, F, H



BFS Traversal

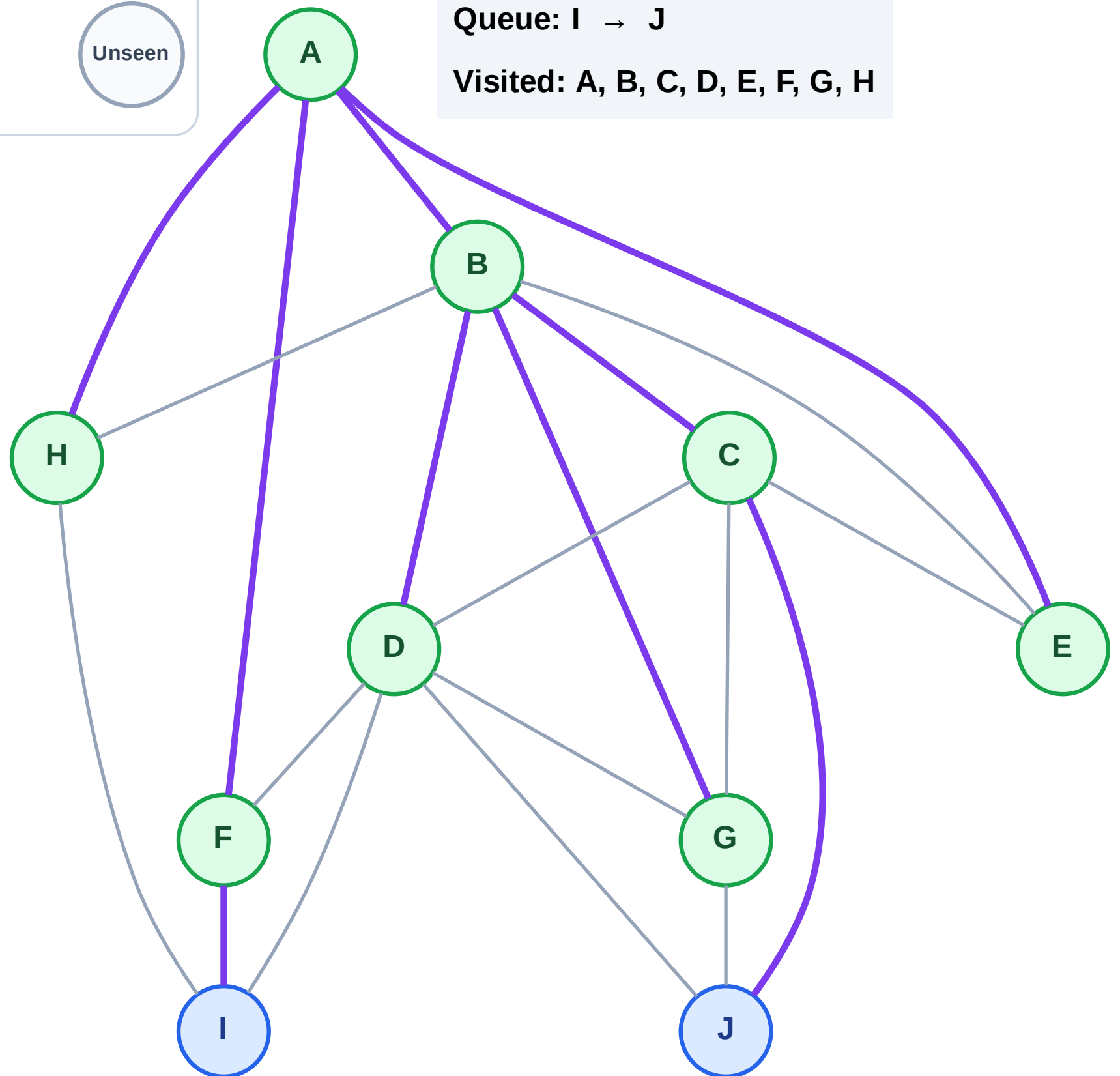
Step 17: No new neighbors from G

Legend



Queue: I → J

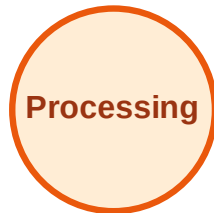
Visited: A, B, C, D, E, F, G, H



BFS Traversal

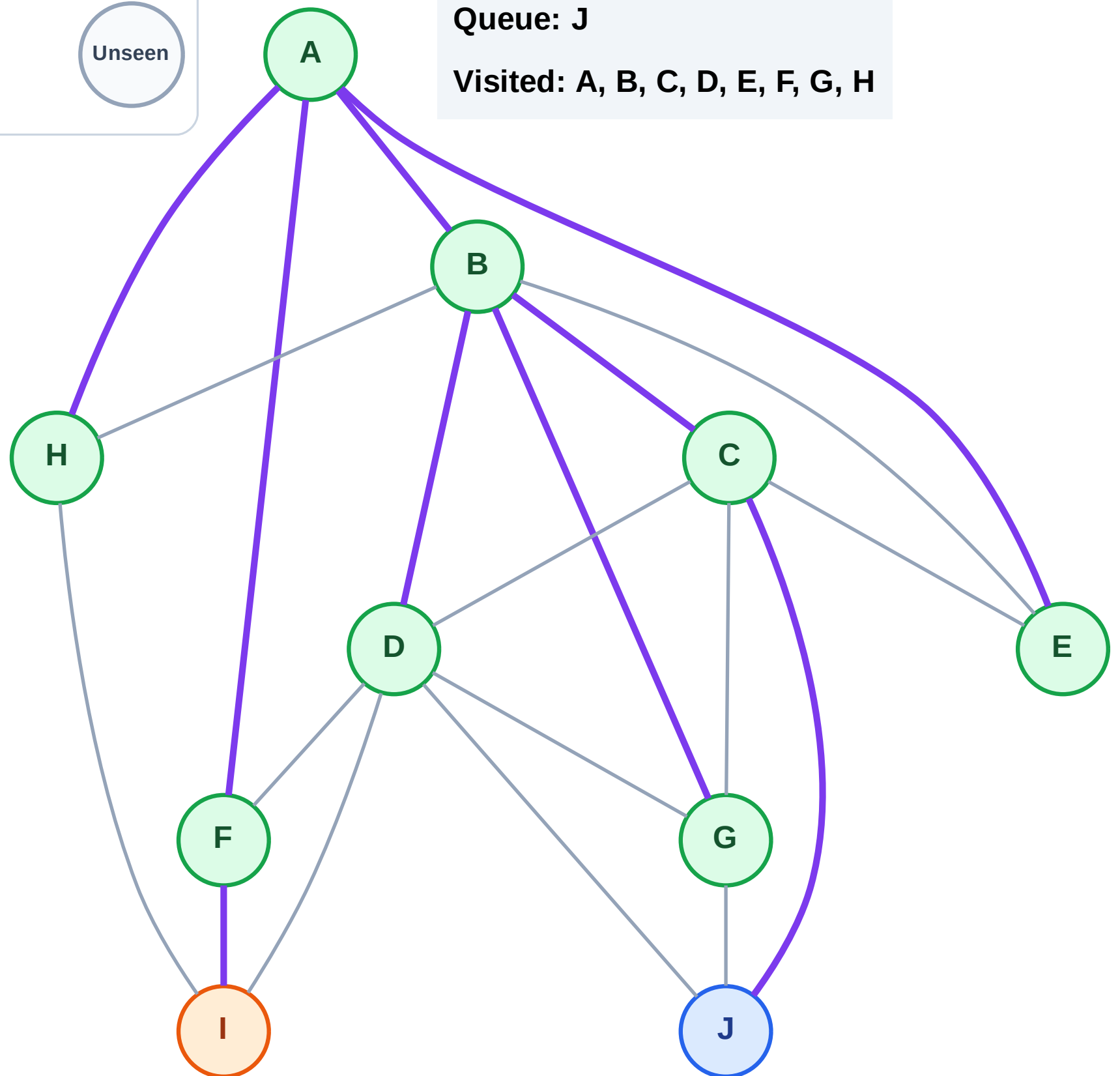
Step 18: Process node I

Legend



Queue: J

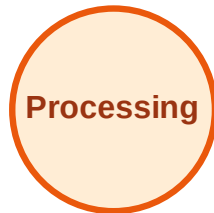
Visited: A, B, C, D, E, F, G, H



BFS Traversal

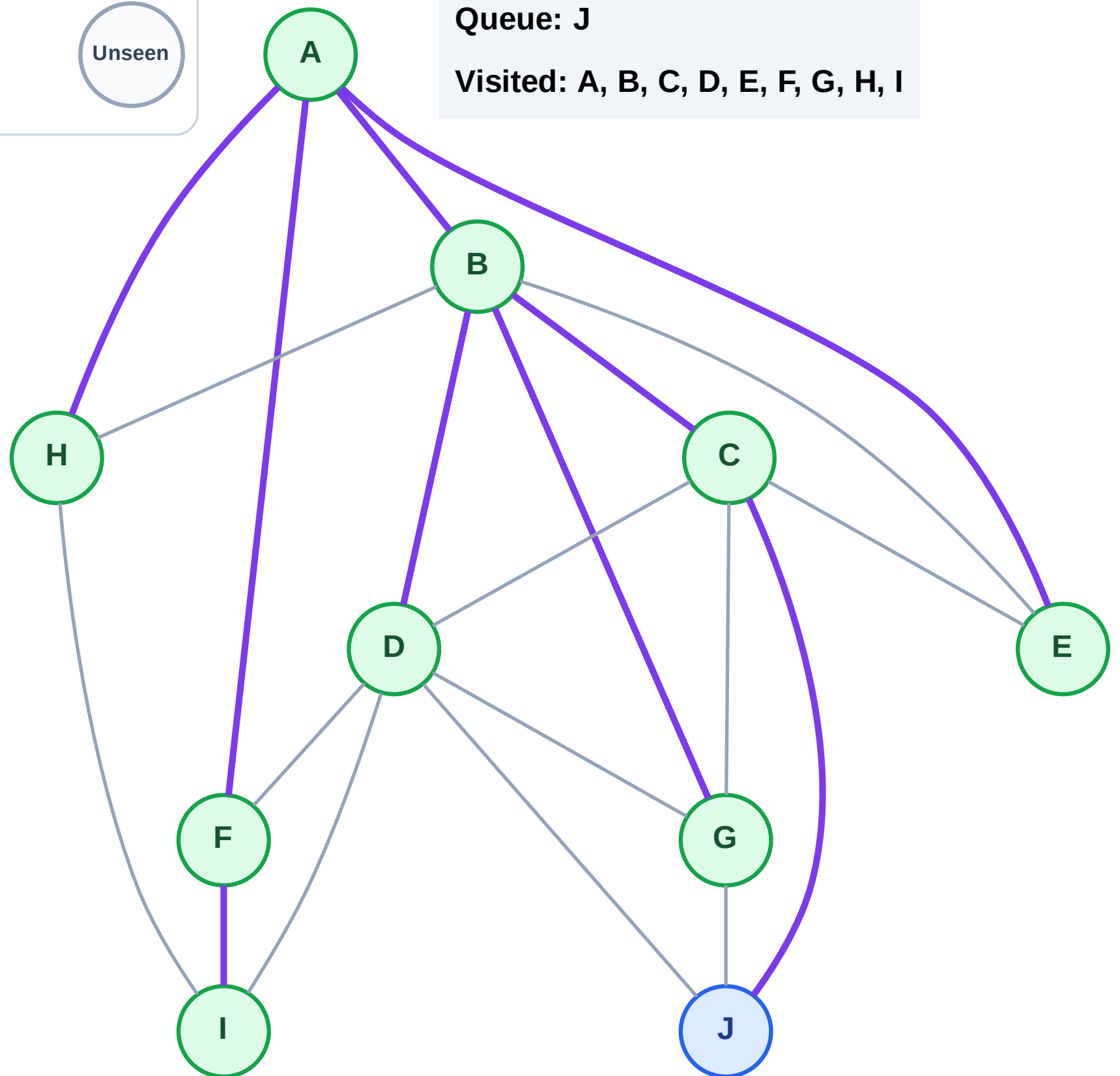
Step 19: No new neighbors from I

Legend



Queue: J

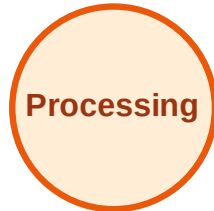
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

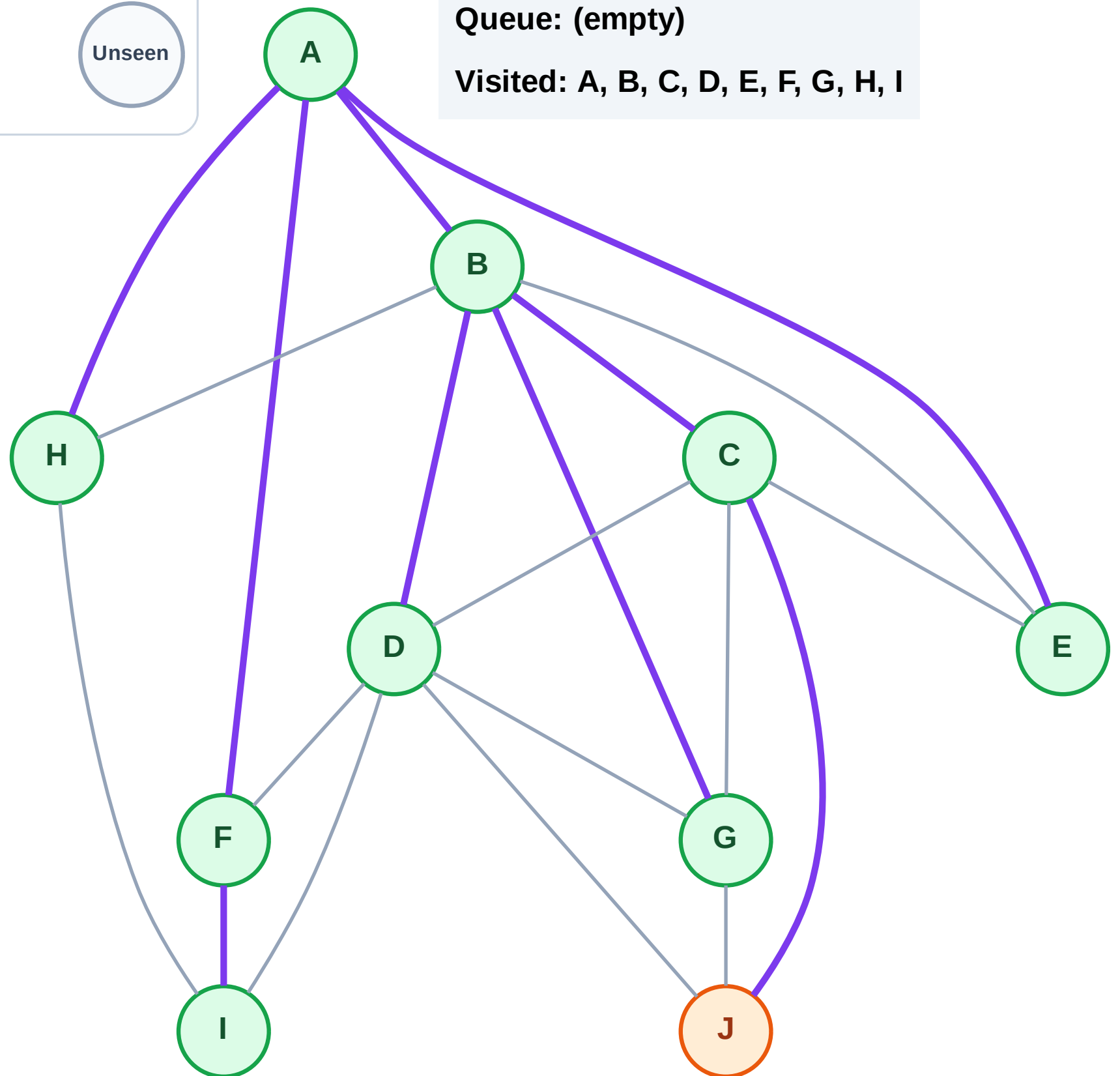
Step 20: Process node J

Legend



Queue: (empty)

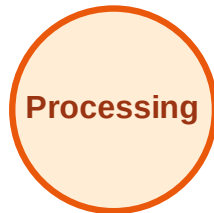
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

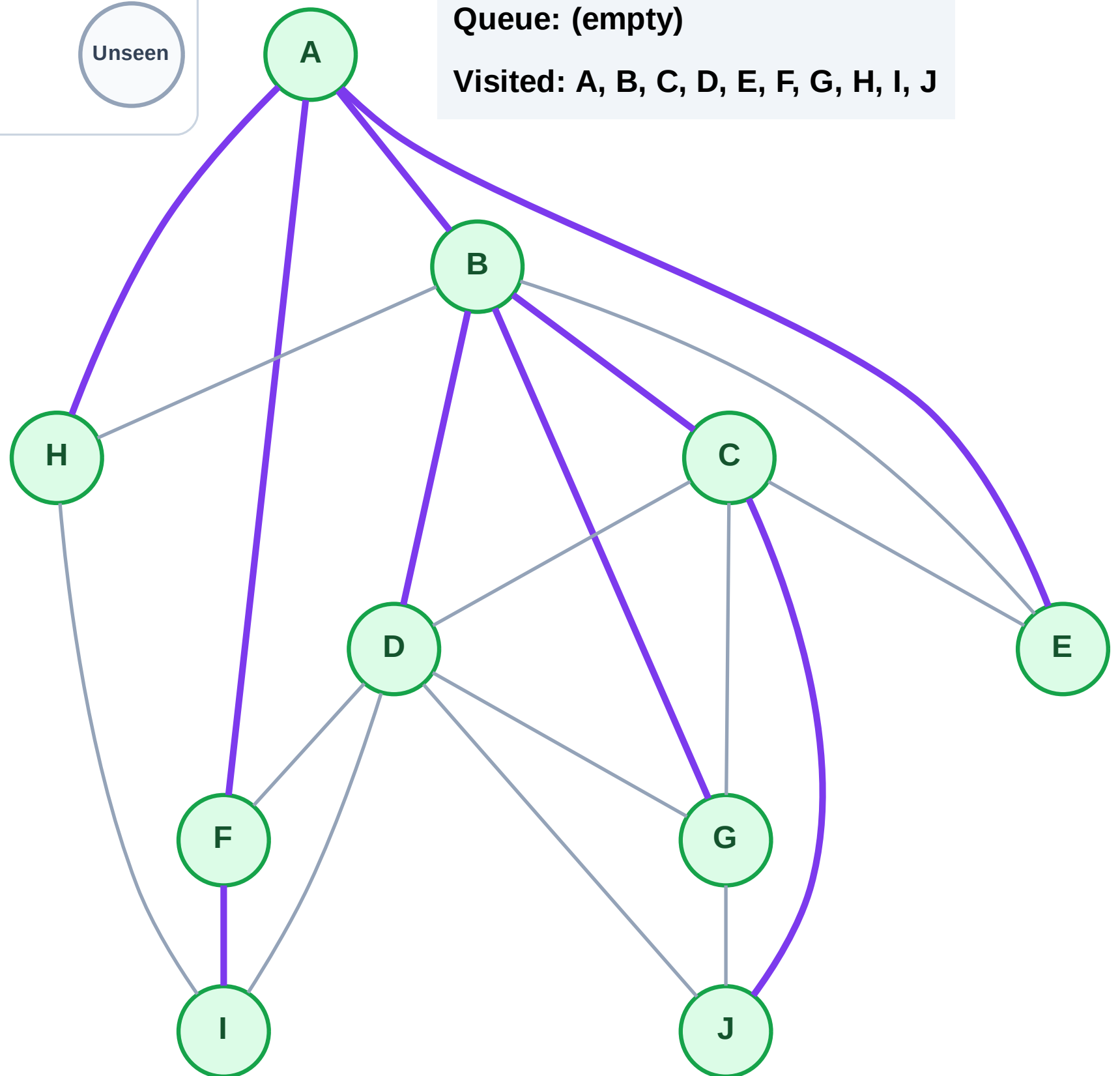
Step 21: No new neighbors from J

Legend



Queue: (empty)

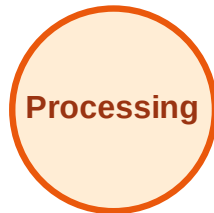
Visited: A, B, C, D, E, F, G, H, I, J



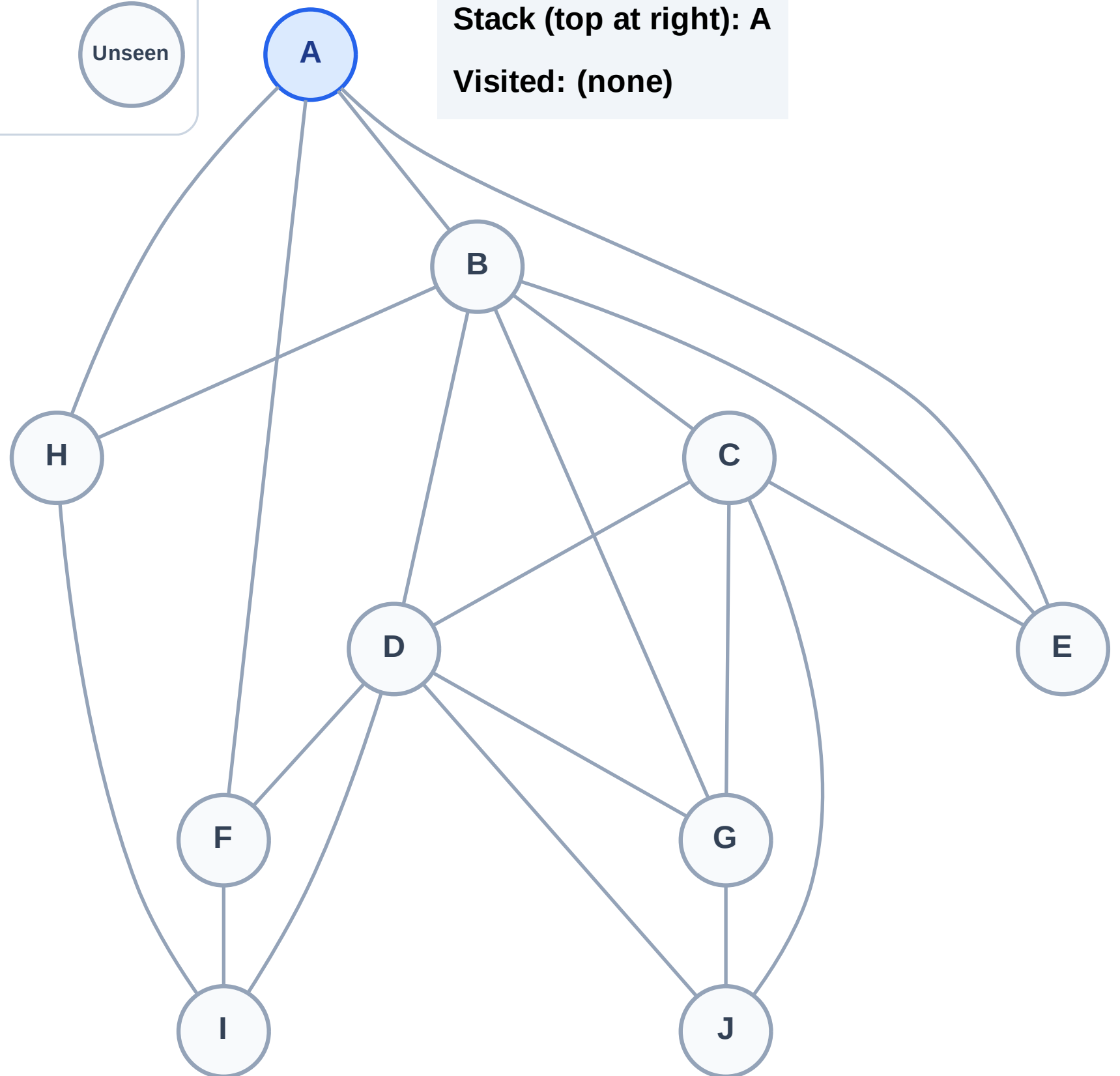
DFS Traversal

Step 1: Start at A

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

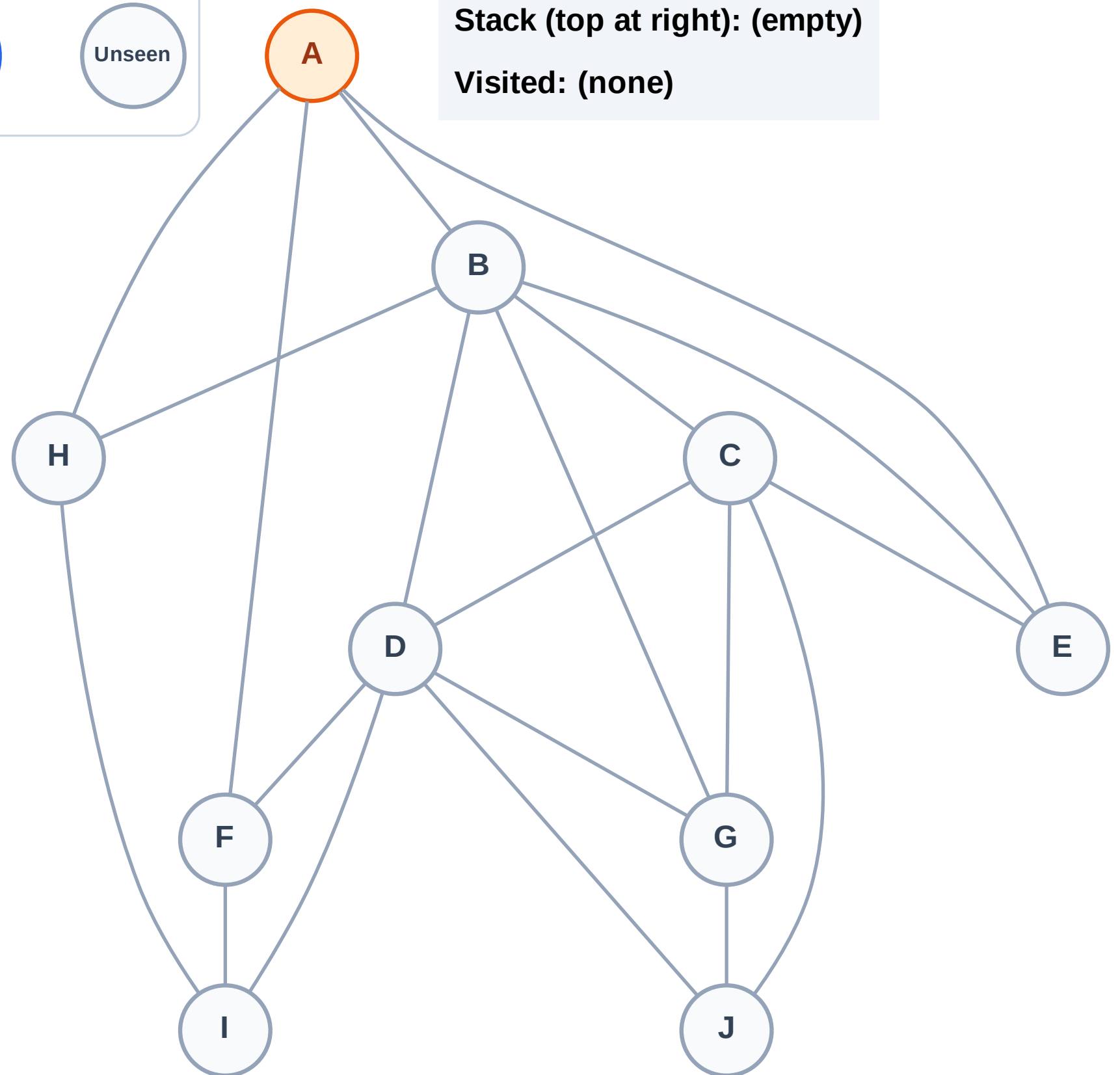
Step 2: Process node A

Legend



Stack (top at right): (empty)

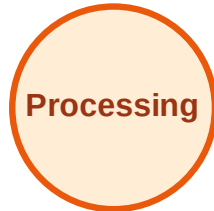
Visited: (none)



DFS Traversal

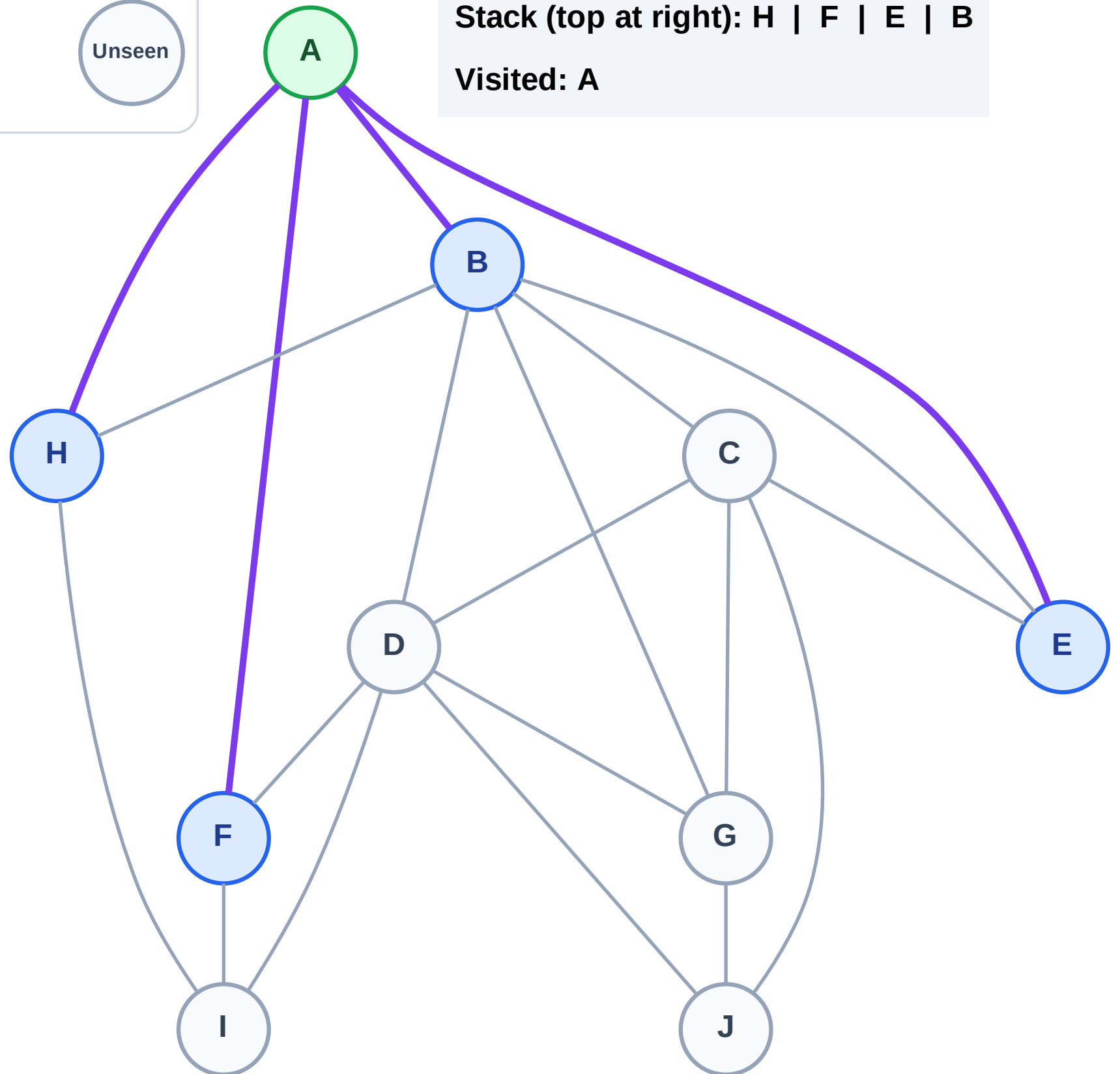
Step 3: Discover H, F, E, B from A

Legend



Stack (top at right): H | F | E | B

Visited: A



DFS Traversal

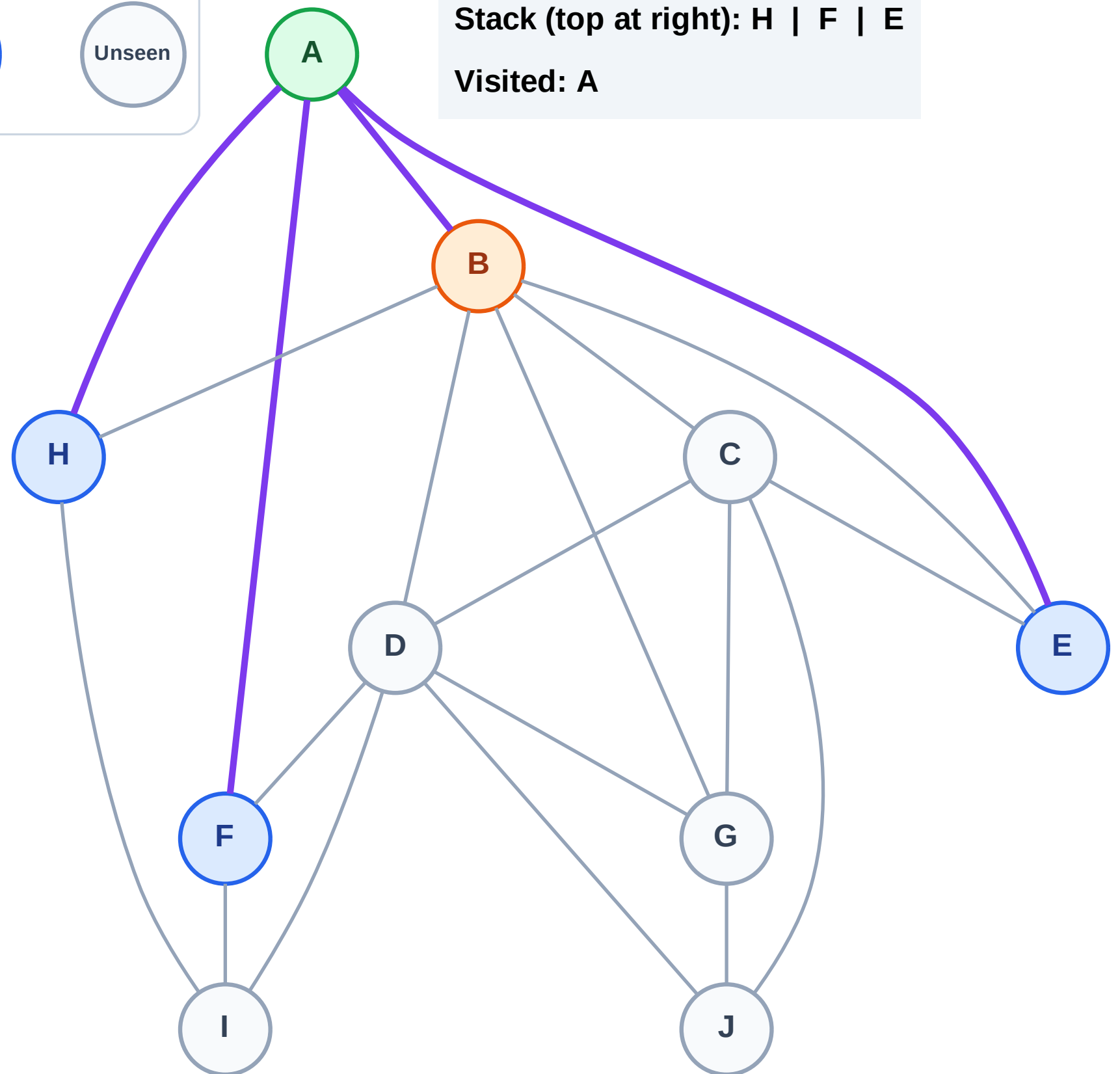
Step 4: Process node B

Legend



Stack (top at right): H | F | E

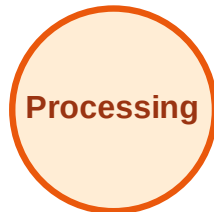
Visited: A



DFS Traversal

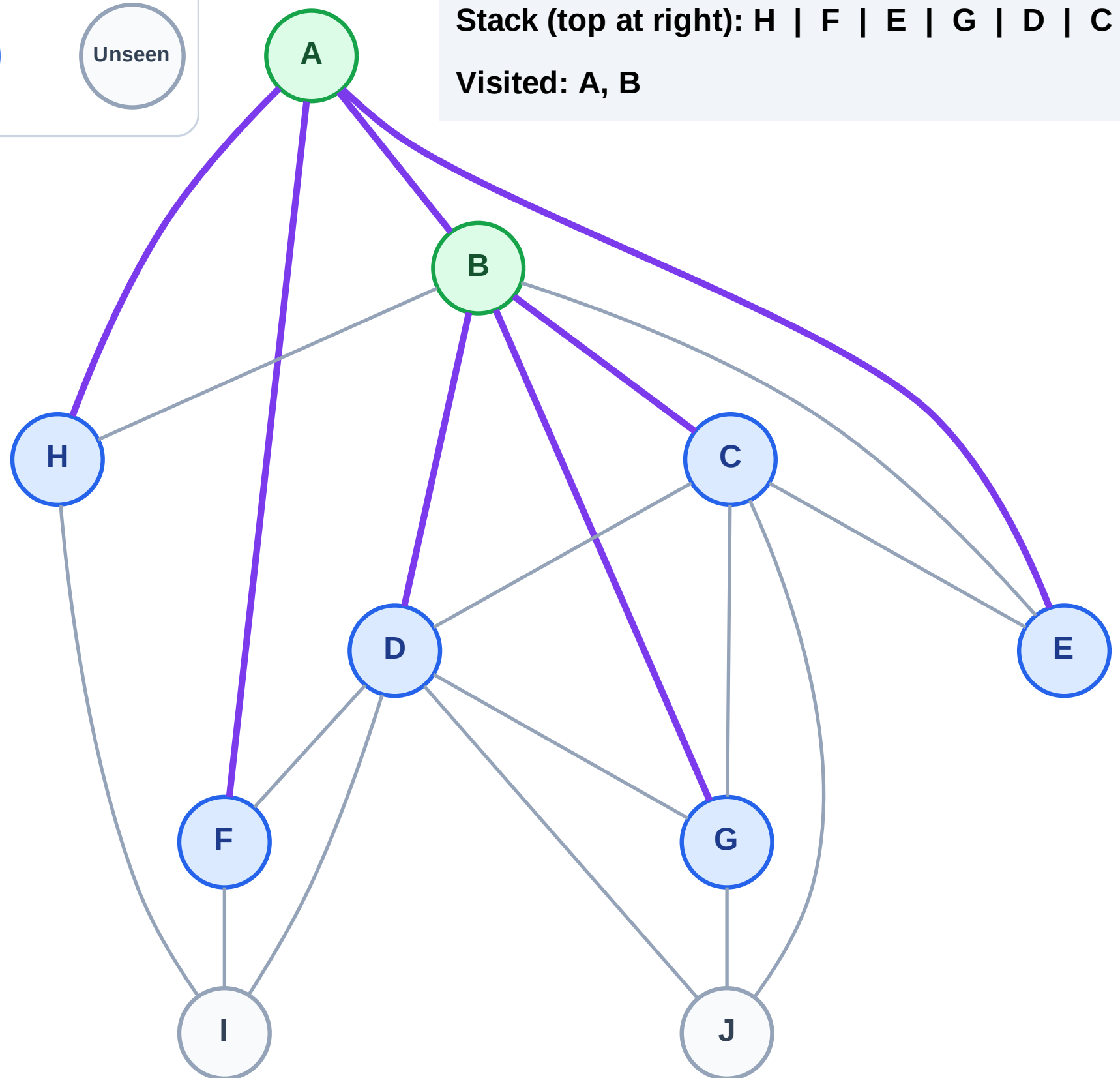
Step 5: Discover G, D, C from B

Legend



Stack (top at right): H | F | E | G | D | C

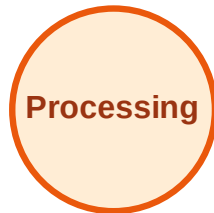
Visited: A, B



DFS Traversal

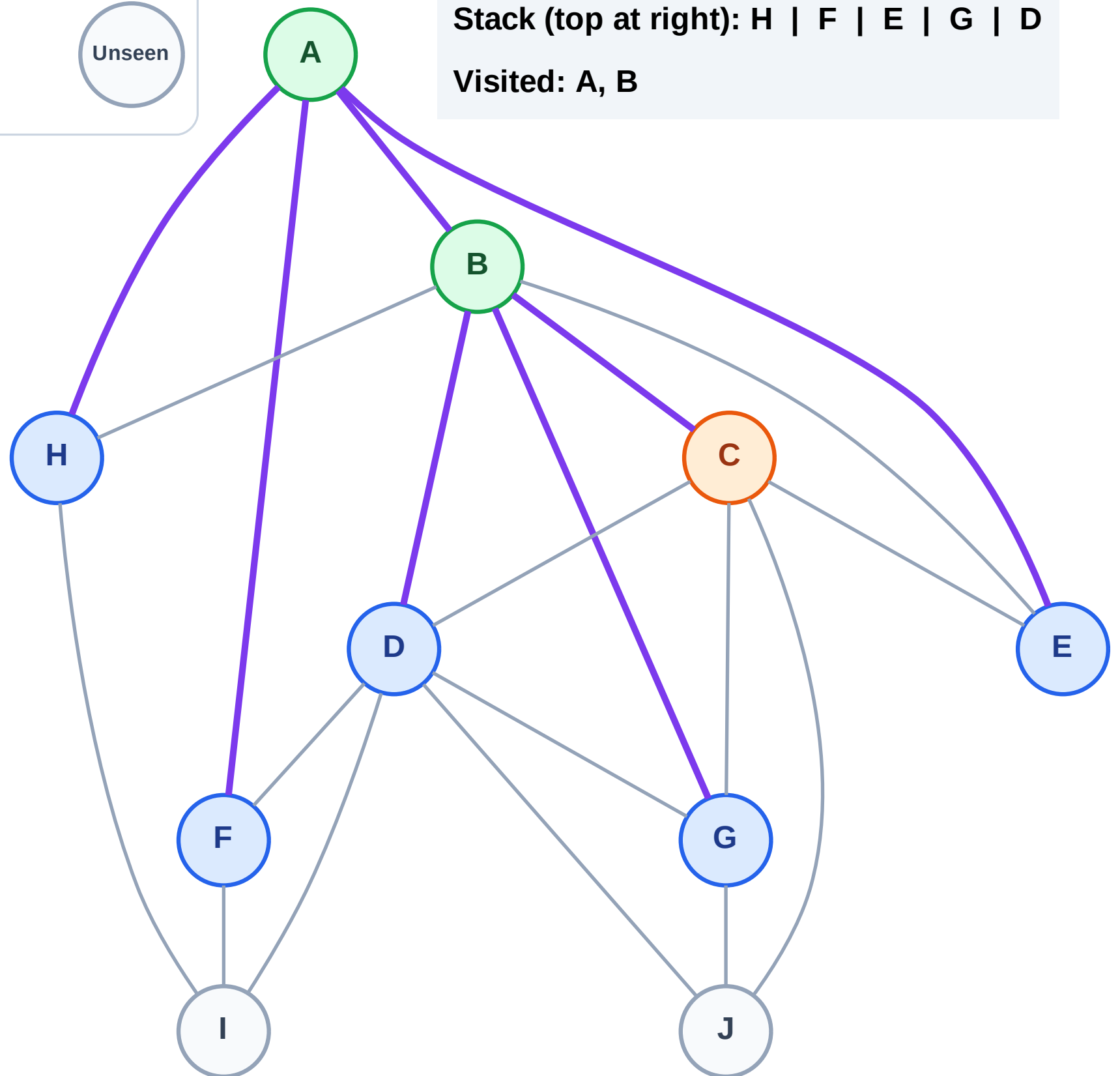
Step 6: Process node C

Legend



Stack (top at right): H | F | E | G | D

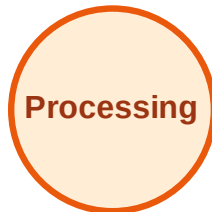
Visited: A, B



DFS Traversal

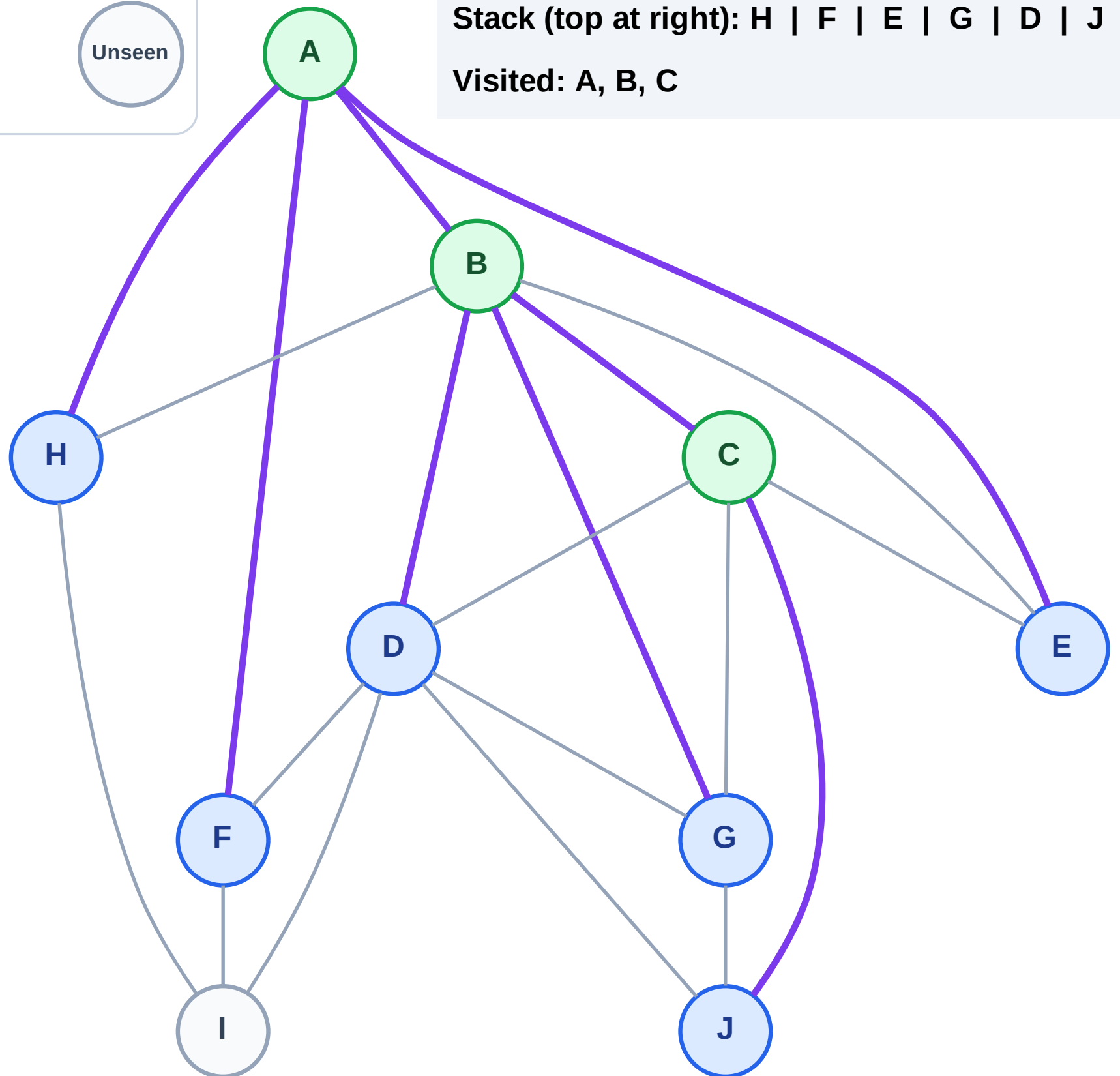
Step 7: Discover J from C

Legend



Stack (top at right): H | F | E | G | D | J

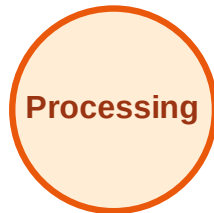
Visited: A, B, C



DFS Traversal

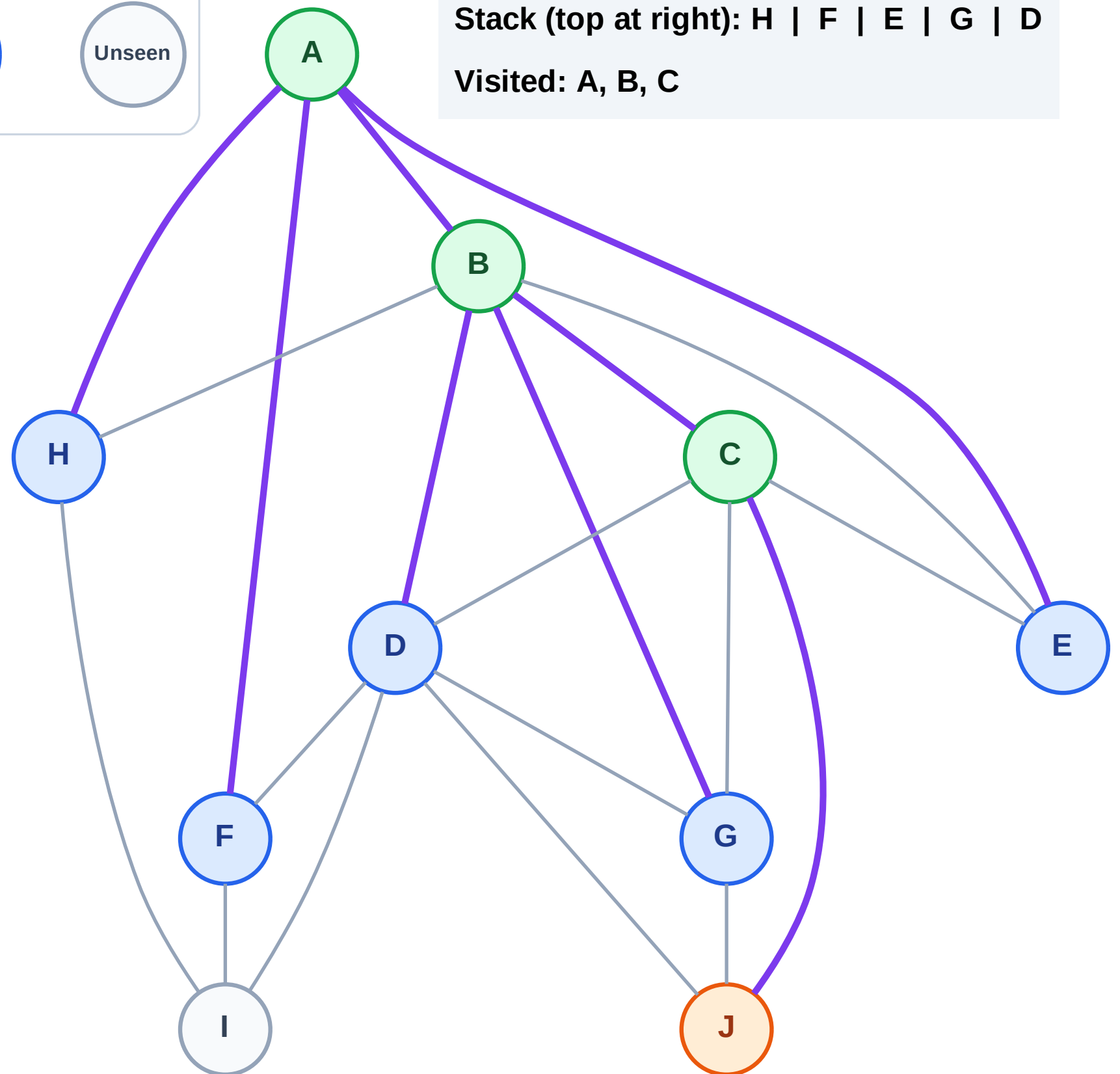
Step 8: Process node J

Legend



Stack (top at right): H | F | E | G | D

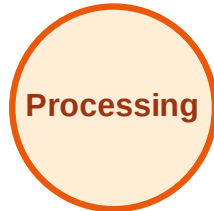
Visited: A, B, C



DFS Traversal

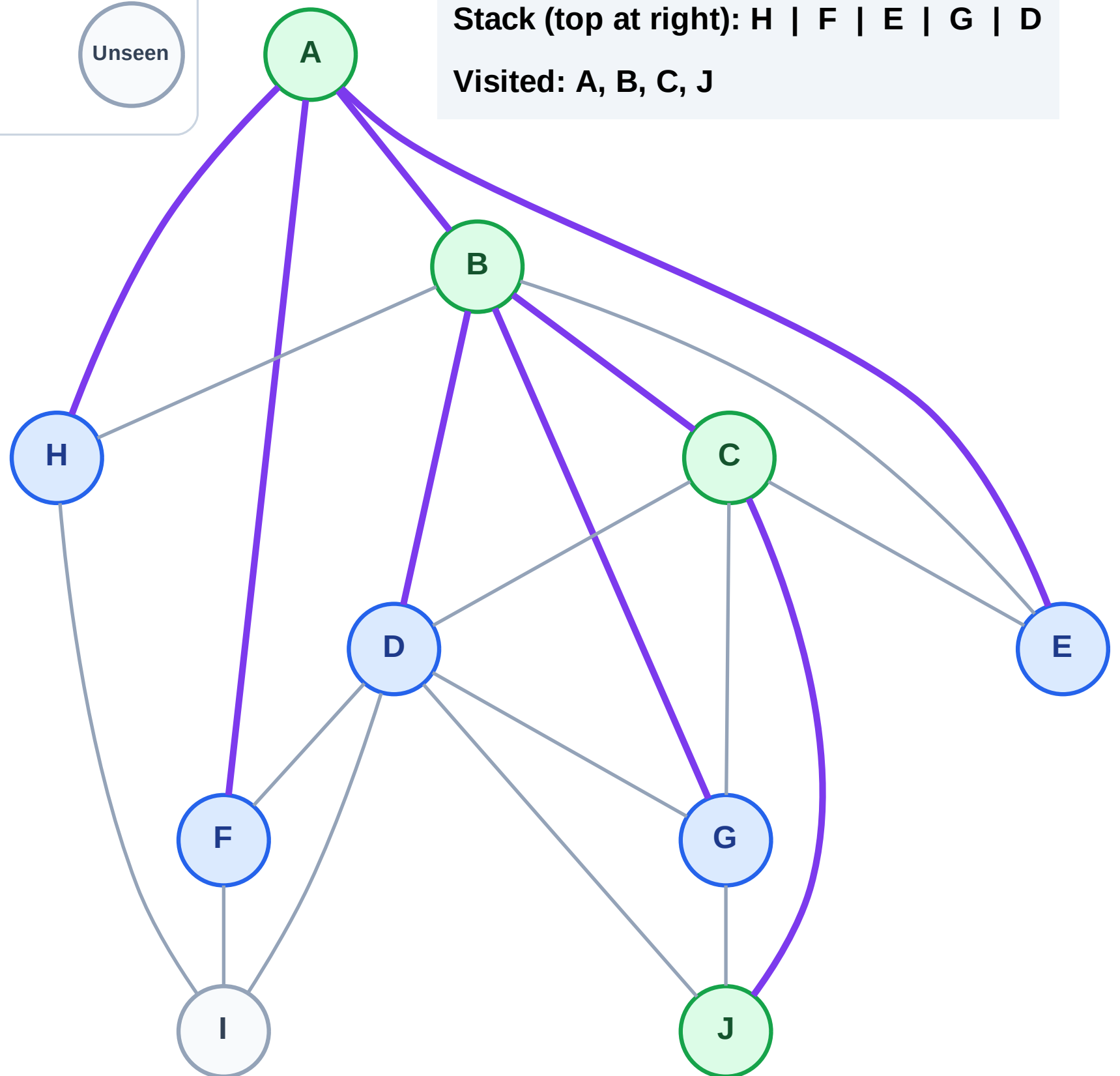
Step 9: No new neighbors from J

Legend



Stack (top at right): H | F | E | G | D

Visited: A, B, C, J



DFS Traversal

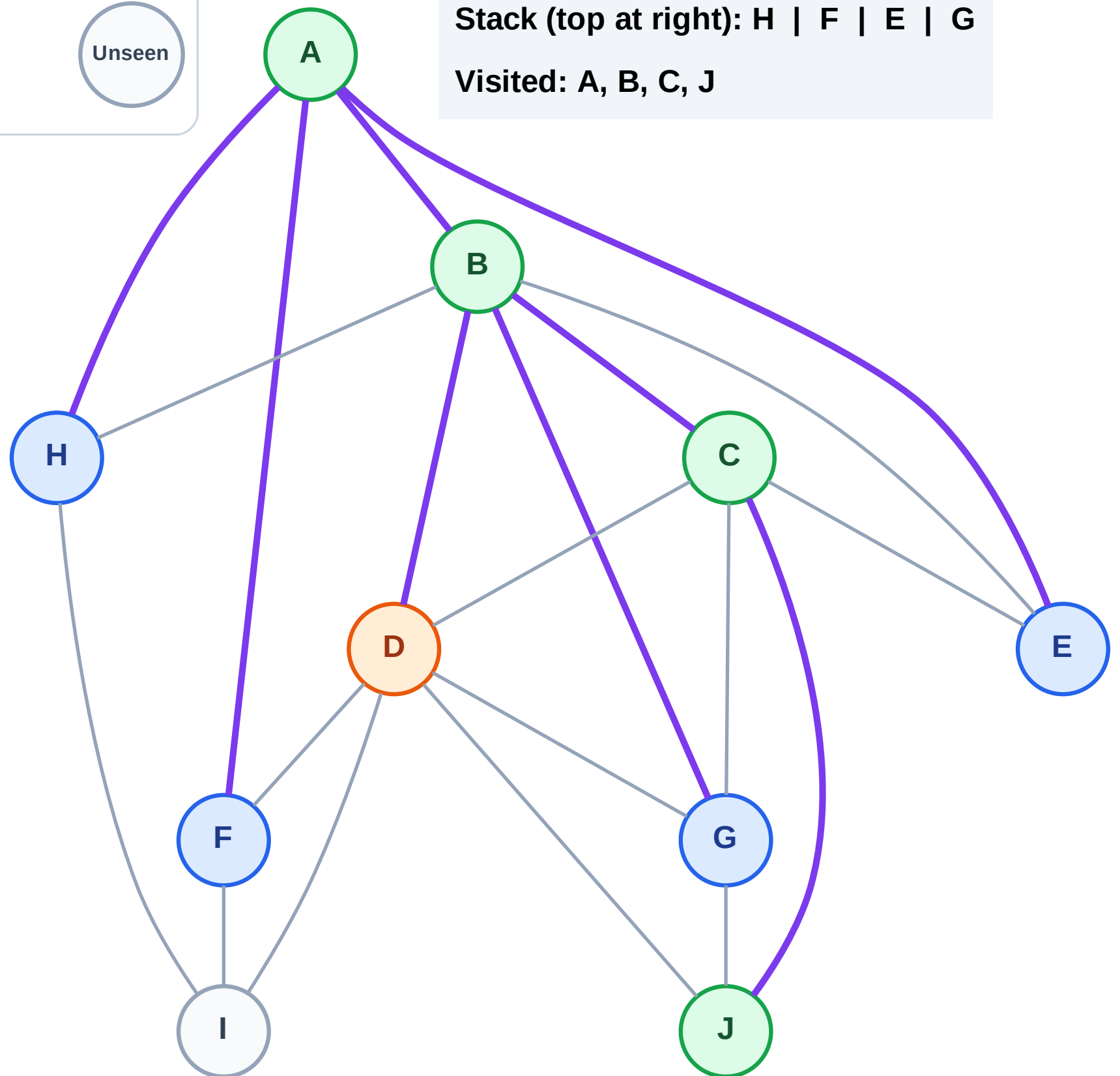
Step 10: Process node D

Legend



Stack (top at right): H | F | E | G

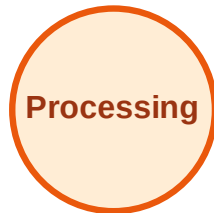
Visited: A, B, C, J



DFS Traversal

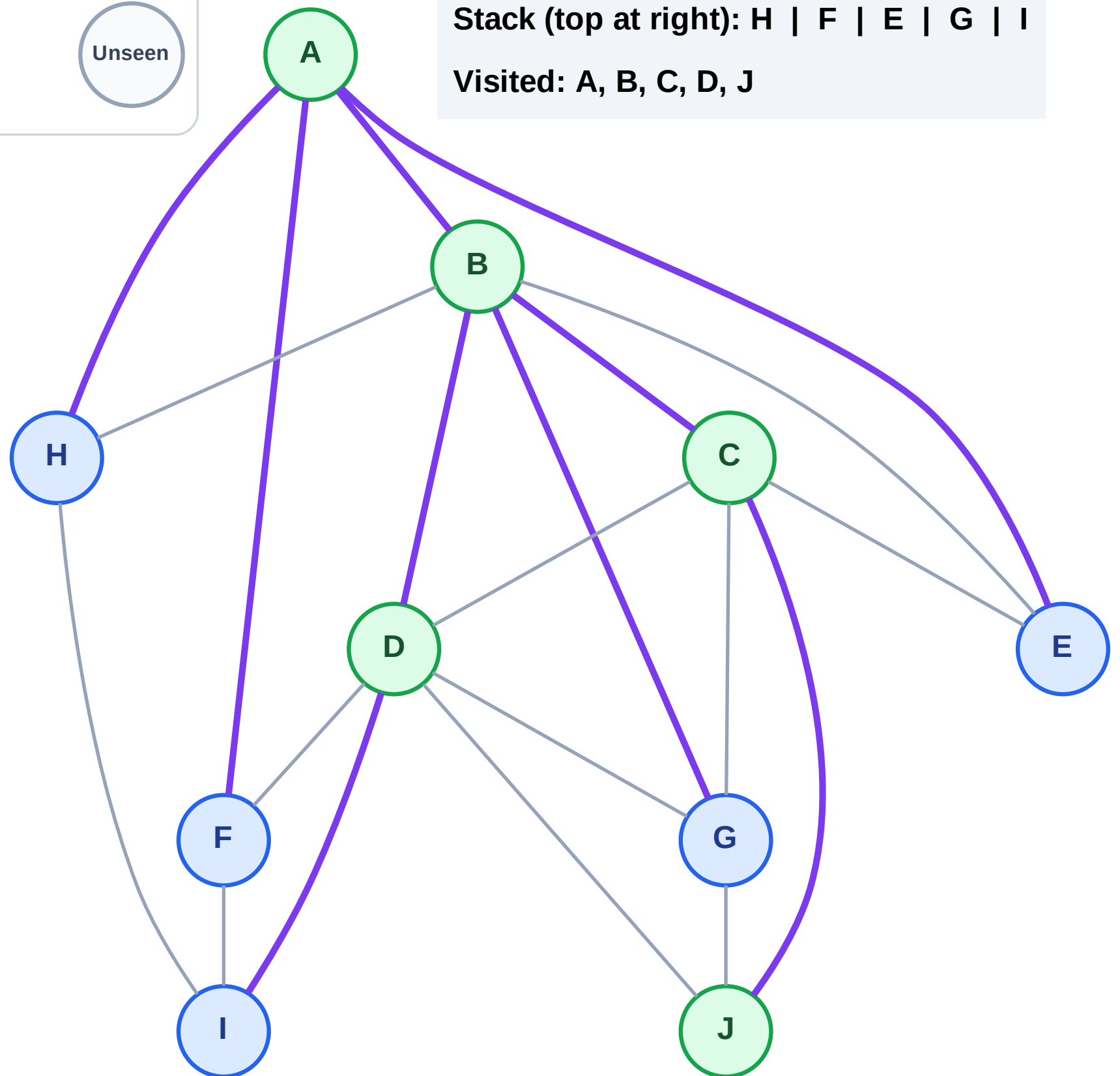
Step 11: Discover I from D

Legend



Stack (top at right): H | F | E | G | I

Visited: A, B, C, D, J



DFS Traversal

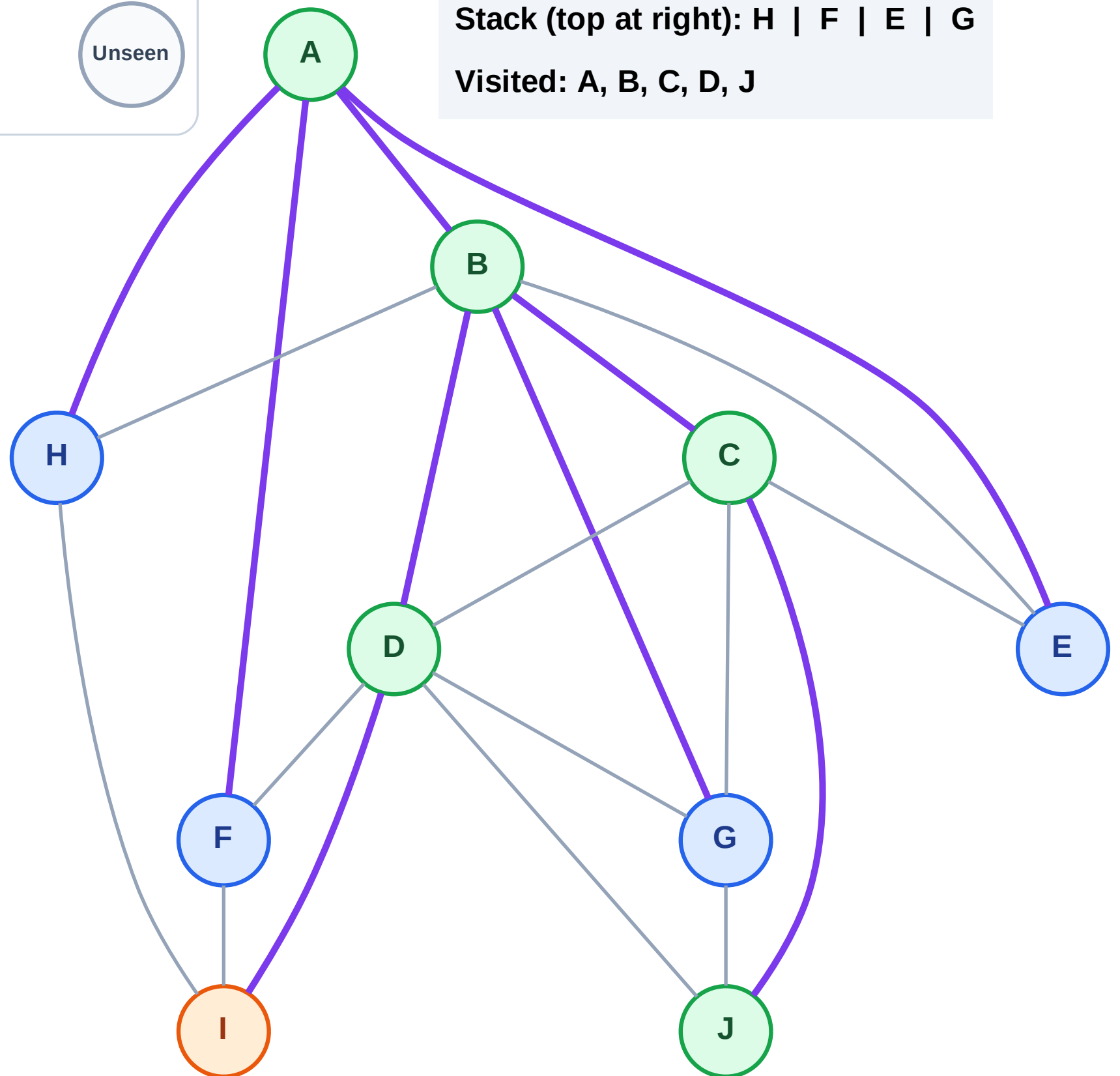
Step 12: Process node I

Legend



Stack (top at right): H | F | E | G

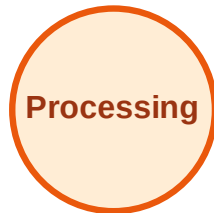
Visited: A, B, C, D, J



DFS Traversal

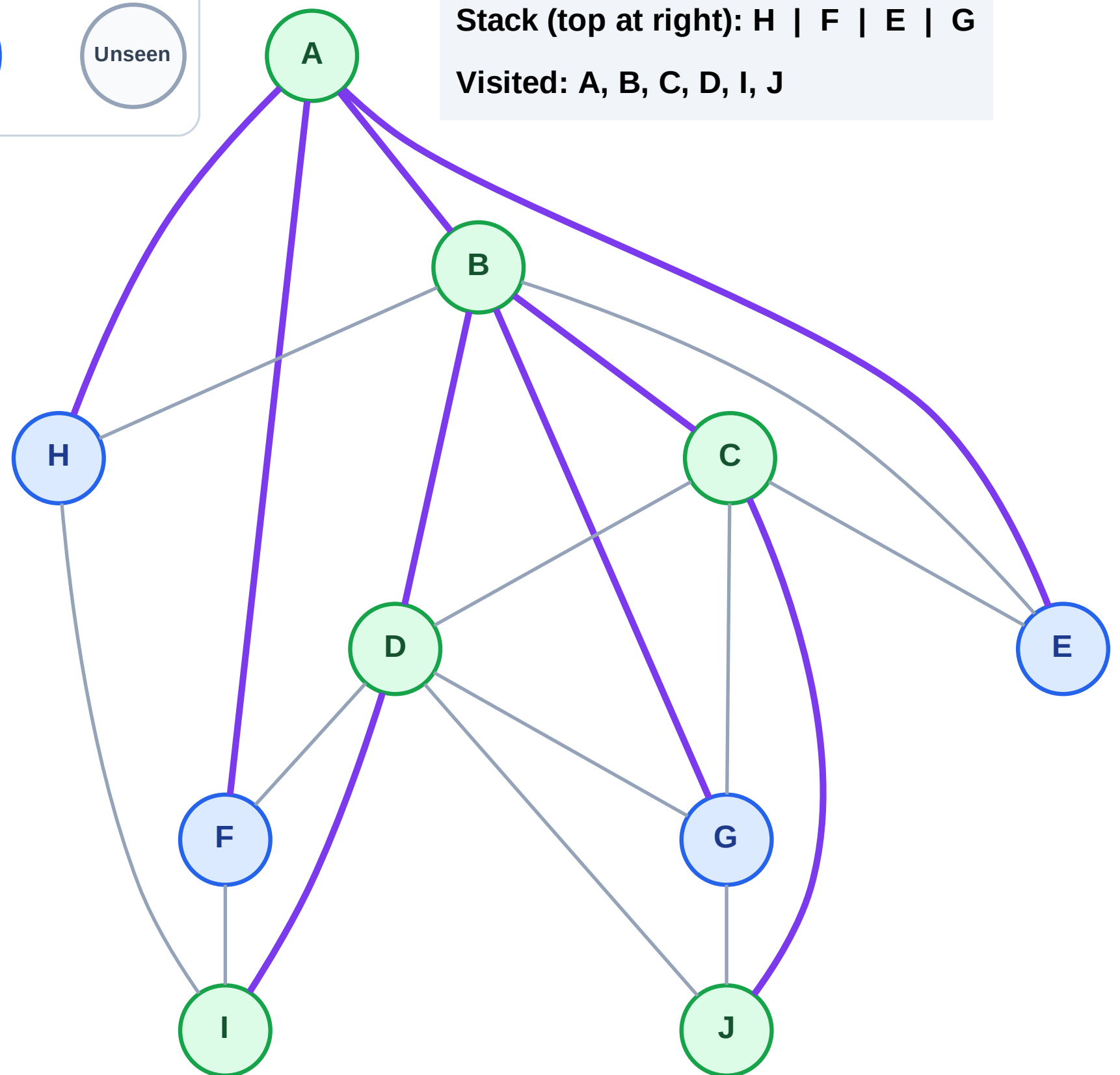
Step 13: No new neighbors from I

Legend



Stack (top at right): H | F | E | G

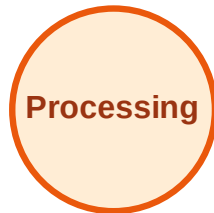
Visited: A, B, C, D, I, J



DFS Traversal

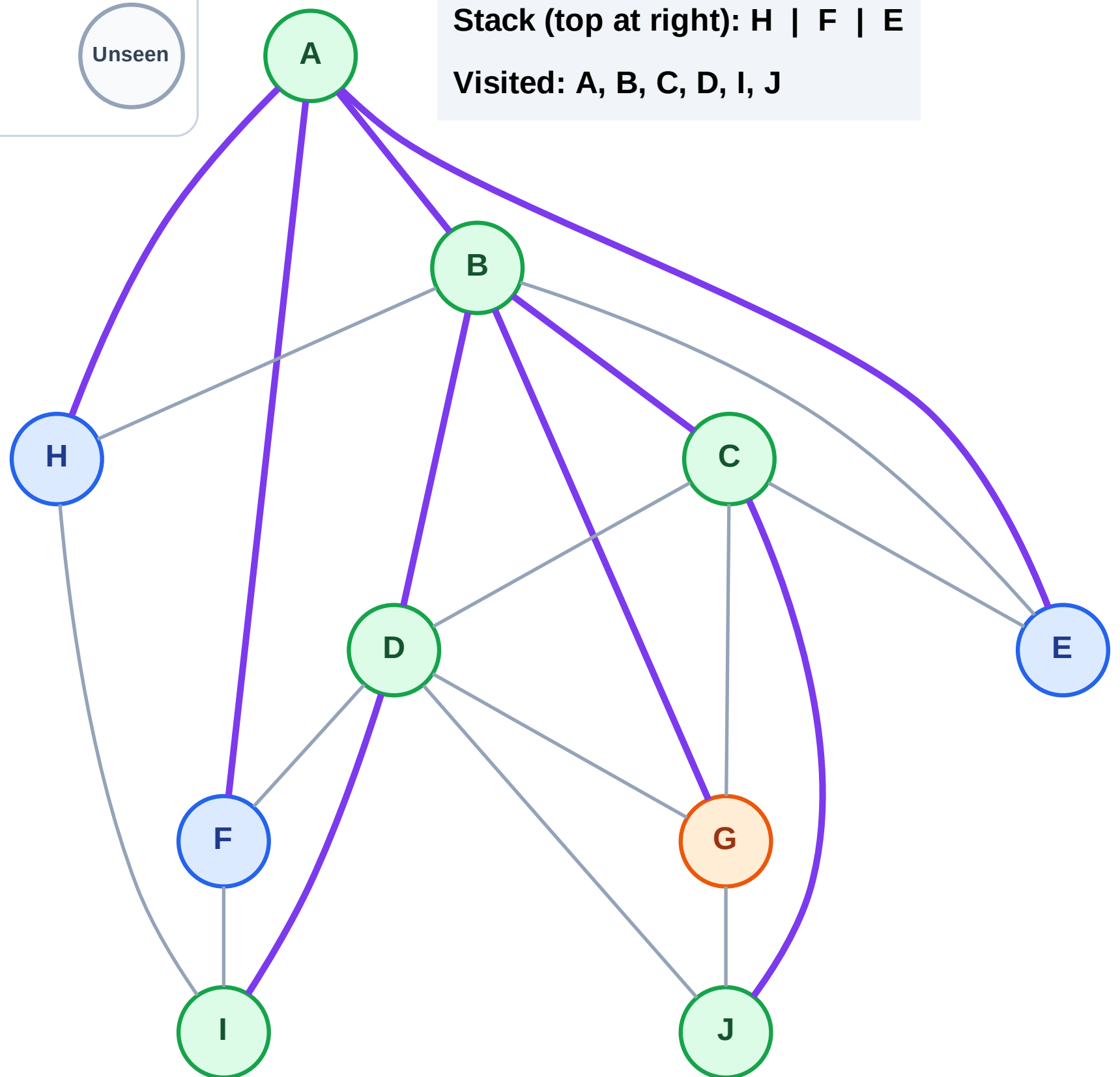
Step 14: Process node G

Legend



Stack (top at right): H | F | E

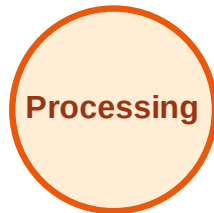
Visited: A, B, C, D, I, J



DFS Traversal

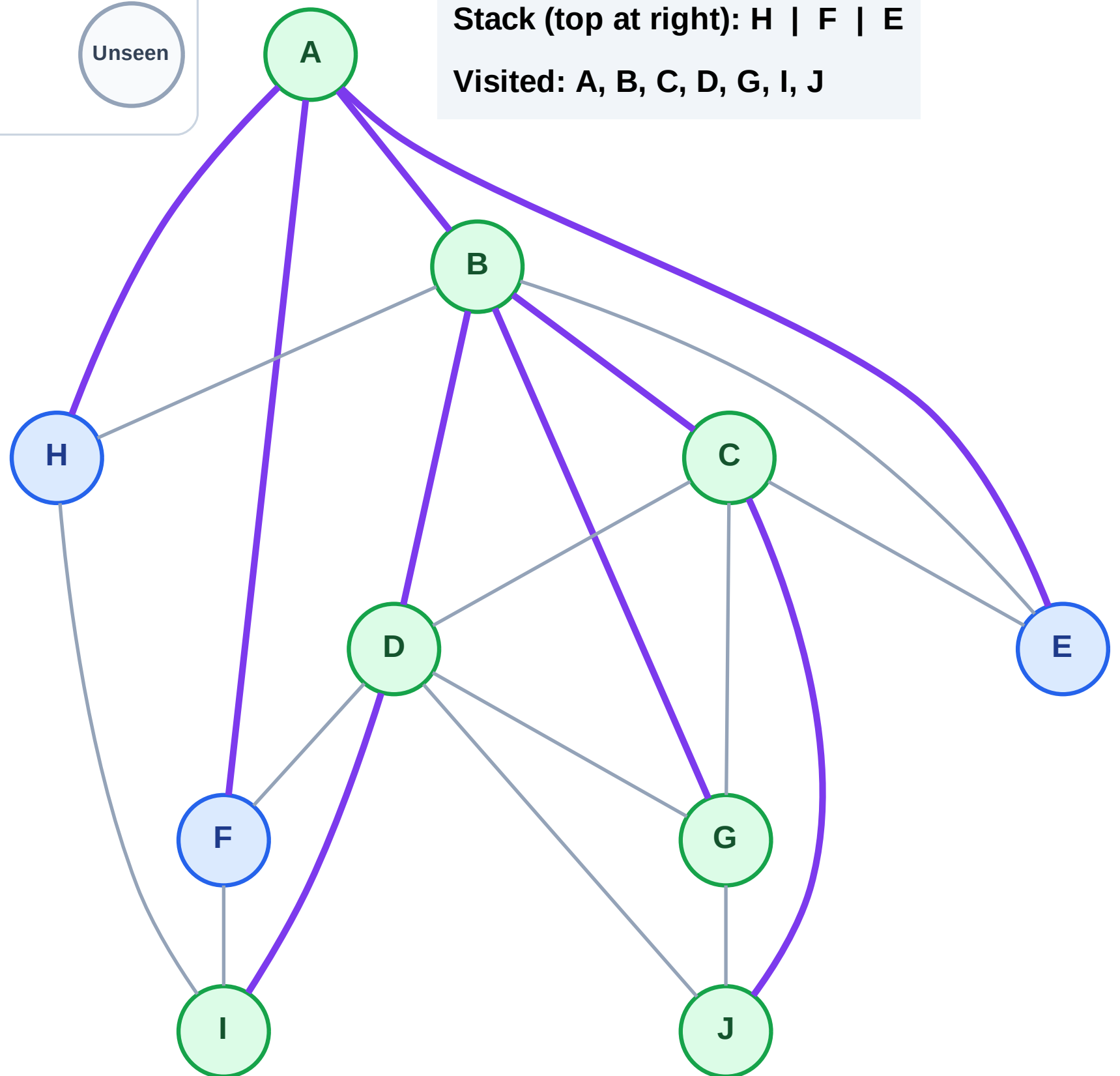
Step 15: No new neighbors from G

Legend



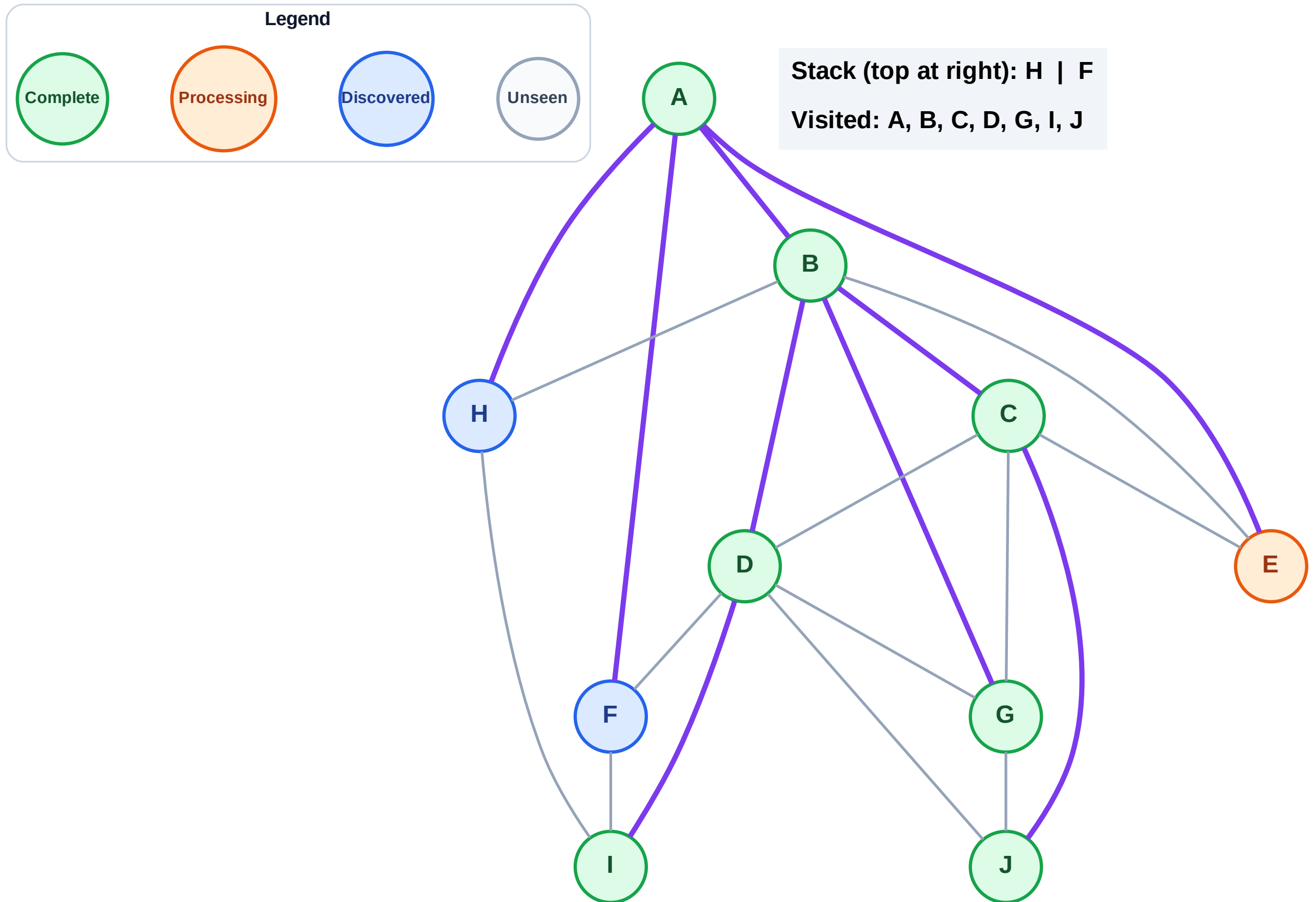
Stack (top at right): H | F | E

Visited: A, B, C, D, G, I, J



Step 16: Process node E

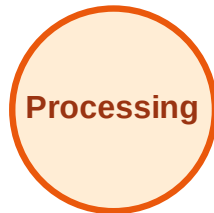
Step 16: Process node E



DFS Traversal

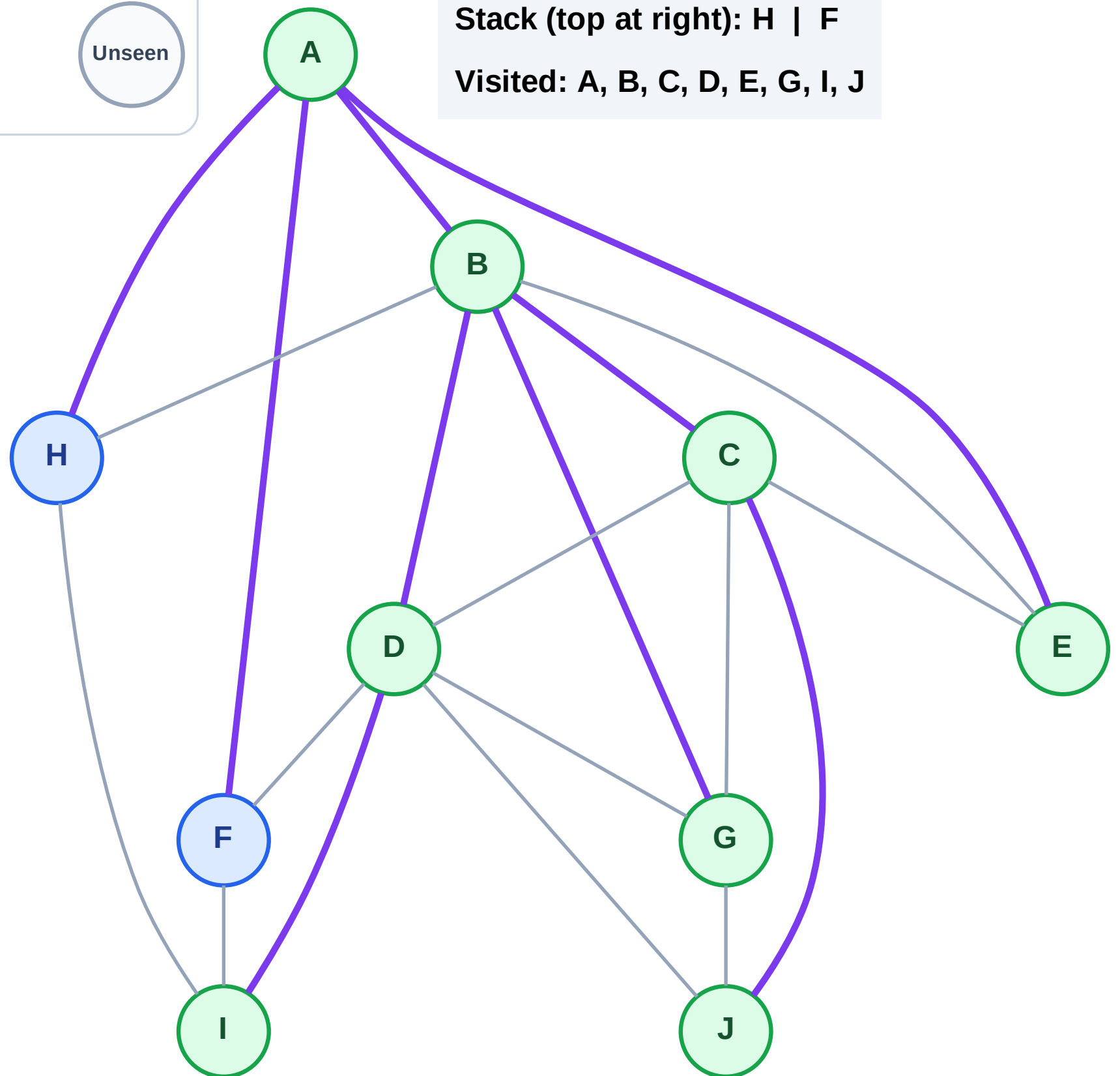
Step 17: No new neighbors from E

Legend



Stack (top at right): H | F

Visited: A, B, C, D, E, G, I, J



DFS Traversal

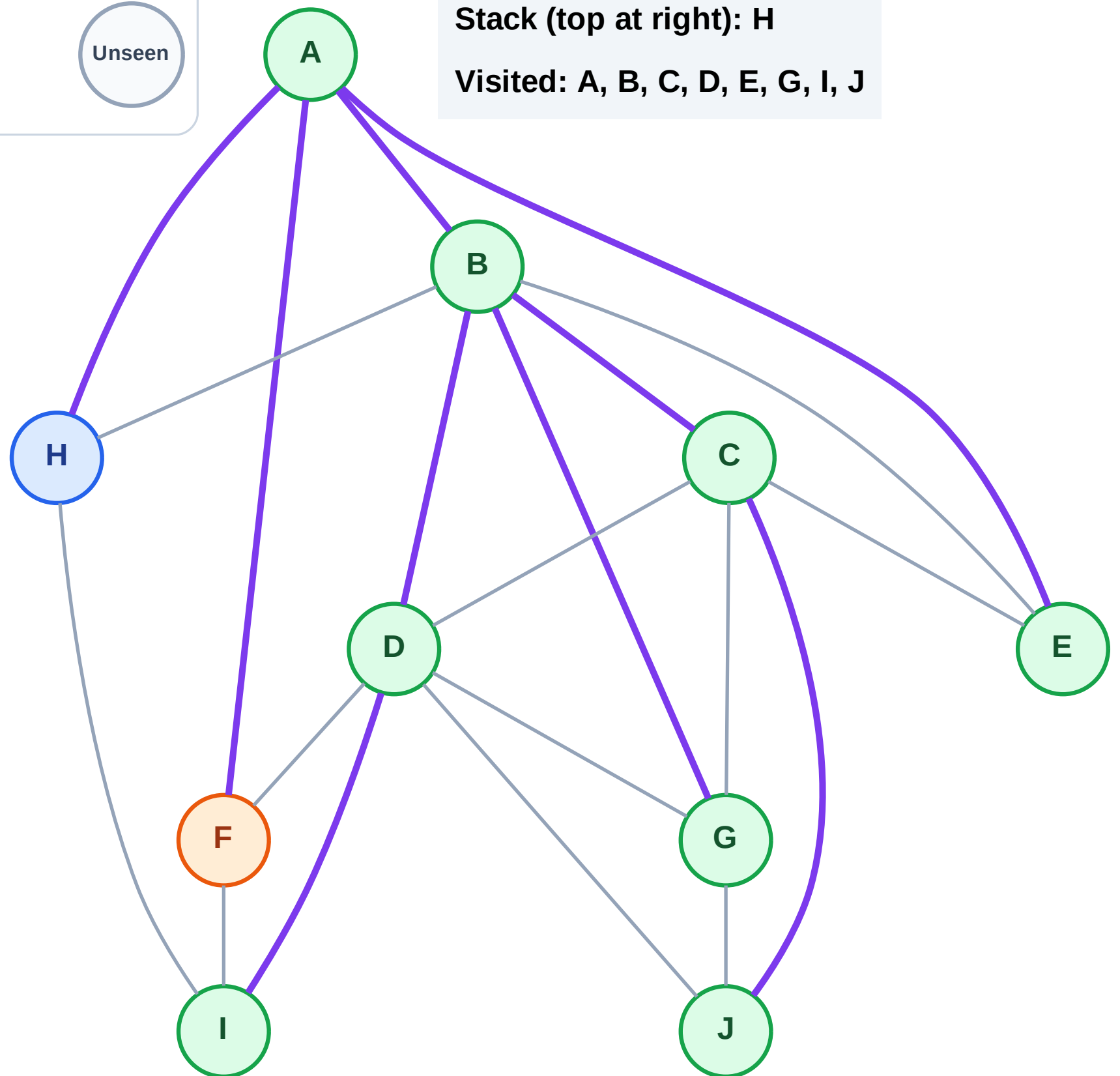
Step 18: Process node F

Legend



Stack (top at right): H

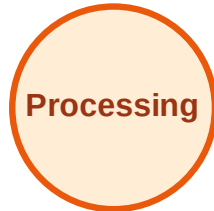
Visited: A, B, C, D, E, G, I, J



DFS Traversal

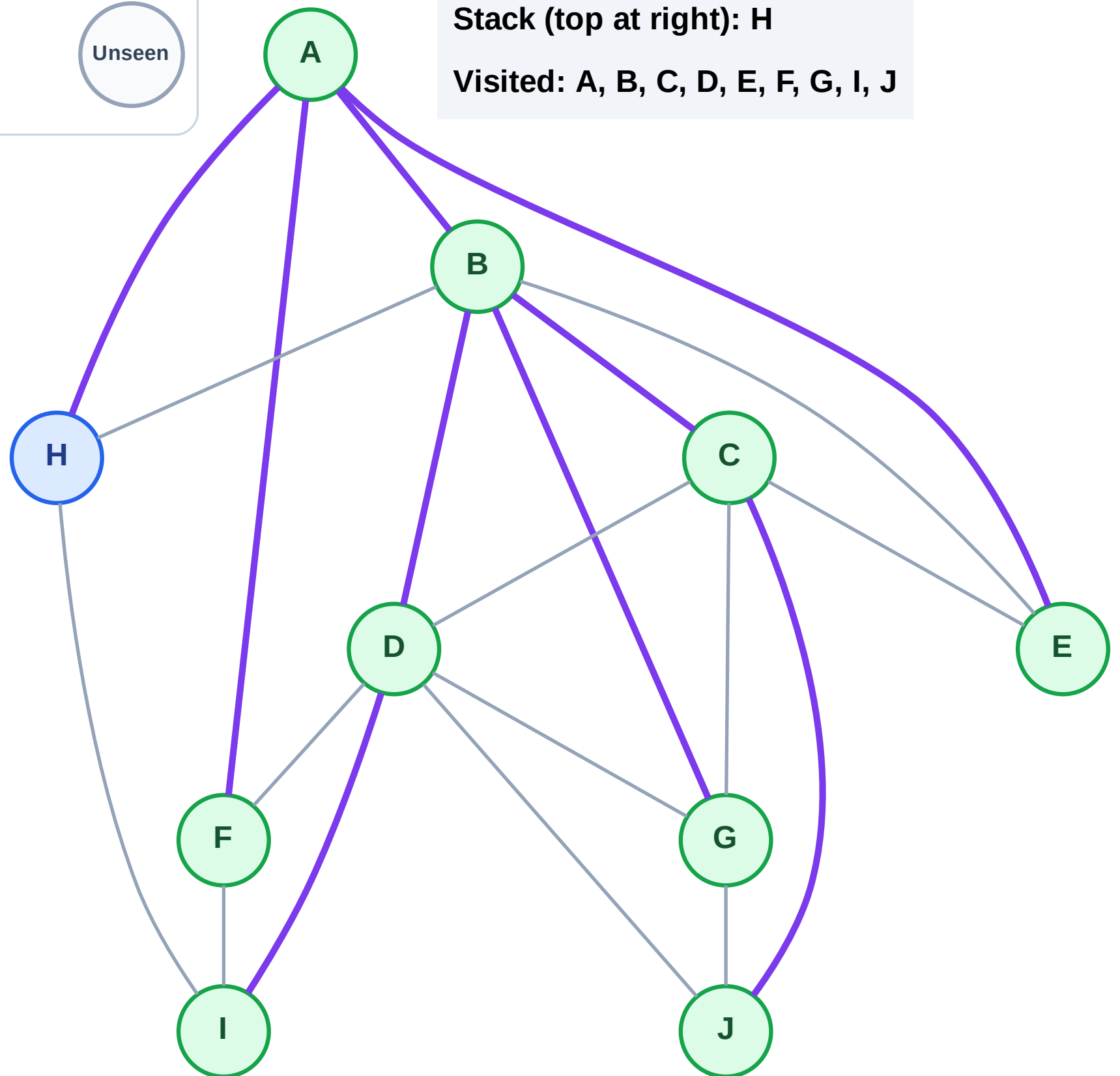
Step 19: No new neighbors from F

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

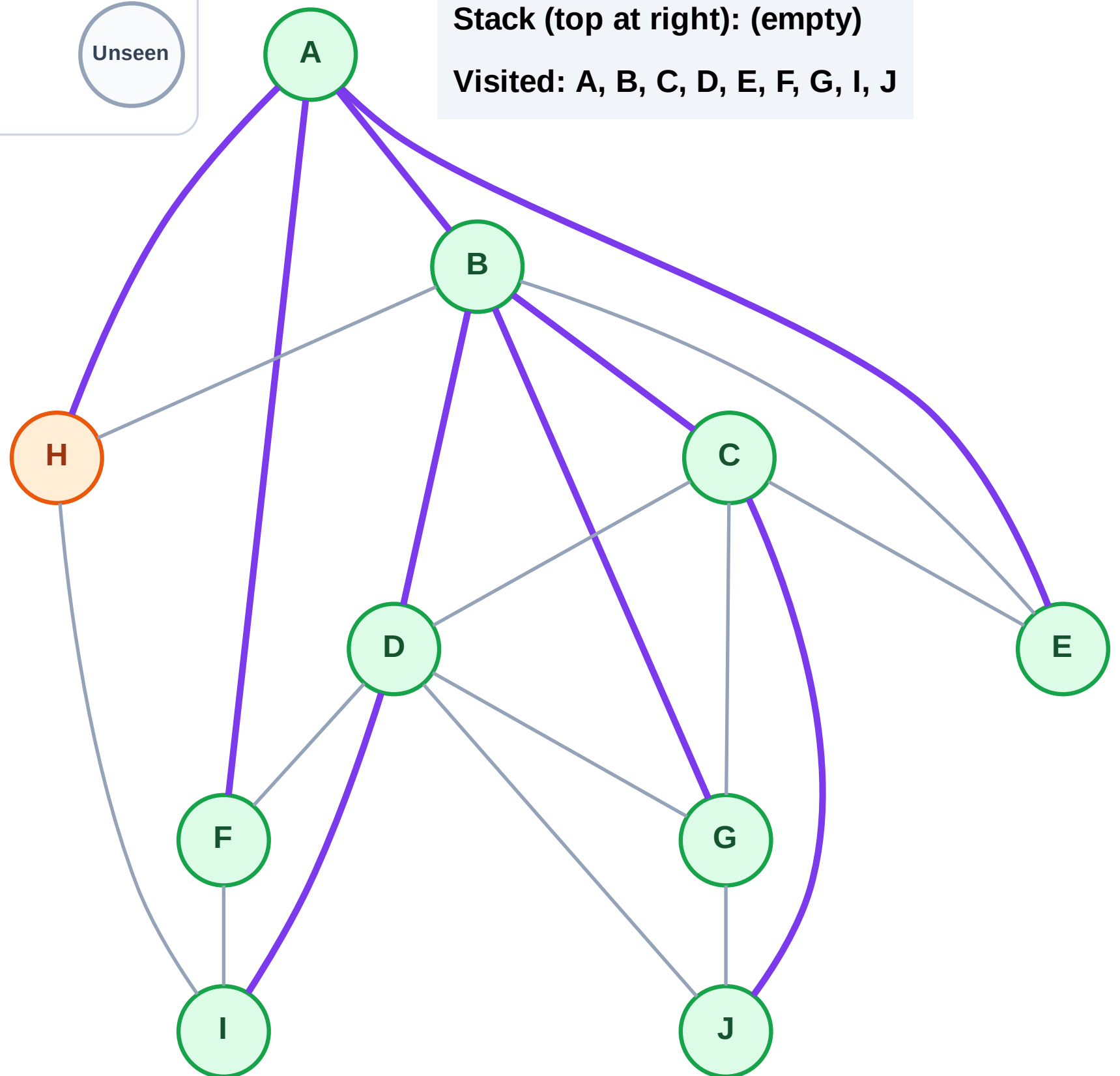
Step 20: Process node H

Legend



Stack (top at right): (empty)

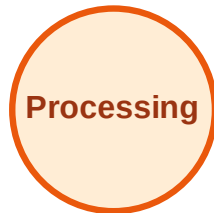
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

